



Sales Tax Automation and Energy-Sector Revenue Projection for the City of Oak Ridge

Summer 2026 Internship Report

Irman Sazzad

MA in Applied Economics

Prepared in connection with an internship administered through

The University of Tennessee Institute for Public Service

Municipal Technical Advisory Service

At City of Oak Ridge, Tennessee

Acknowledgements

I would like to express my sincere gratitude to the University of Tennessee Institute for Public Service and the Municipal Technical Advisory Service for providing me with the opportunity to participate in this internship and gain meaningful experience within Tennessee local government. As an international student seeking to connect a rigorous background in quantitative economics with applied public policy, this opportunity was especially significant. IPS created a platform through which I could learn how local government functions, contribute to substantive public-sector assignments, and develop confidence in my ability to pursue a career involving public policy, economic analysis, and community service.

I am deeply grateful to Dr. Herb Byrd, Vice President of Public Service, Ms. Angie Carrier, MTAS Executive Director, Mr. Jon Walden, CTAS Executive Director, Ms. Kristen Davis, Director of Development, Ms. Margaret Norris, Municipal Management Consultant, Ms. Tomi Rogers, HR and Operations Manager, Ms. Valerie Dugan, Administrative Specialist, Ms. Susan Robertson, Communications Manager, and the other IPS and MTAS professionals who supported the internship program, coordinated opportunities, provided guidance, and made interns feel welcomed and supported throughout the experience. Their work created an environment in which students were trusted with meaningful responsibilities and encouraged to learn directly from public officials and municipal professionals.

I extend my deepest appreciation to Tom Pessemier, Deputy City Manager of the City of Oak Ridge, for serving as my primary supervisor. His guidance, trust, patience, and encouragement were essential to the completion of both internship assignments. He allowed me the freedom to investigate difficult problems, test different approaches, learn through trial and error, and develop solutions that went beyond the initial project expectations. His willingness to discuss technical challenges, explain the broader municipal importance of the work, and include me in meetings with senior officials made the internship an exceptional professional experience.

I am also deeply grateful to Mr. Randy Hemann, City Manager; Ms. Kendahl Moore, Assistant City Manager; Ms. Kari Bates, Finance Director; Dr. Amy Fitzgerald, Director of Government Affairs; Ms. MacKenzie Hetrick, Executive Associate in the City Manager's Office; Ms. Lauren Gray, Senior Communications Specialist; Mr. Eric Ault, Personnel Director and every other staff member and official at the City of Oak Ridge who guided, supported, encouraged, and inspired me throughout the internship. Their willingness to share their time, expertise, and institutional knowledge helped me move forward whenever I faced challenges and played an important role in my professional growth.

I would also like to recognize the incredible police officers, firefighters, emergency personnel, elected officials, public works staff, planners, utility employees who I encountered during the internship. Their dedication, readiness, professionalism, humor, and pride in serving the community profoundly changed my understanding of local government. Much of their work is performed quietly and without public recognition, yet it is essential to keeping the City safe, functional, financially responsible, and prepared for future growth. I remain deeply grateful to everyone at the City of Oak Ridge for welcoming me into an institution with such extraordinary historical, scientific, economic, and public-service significance.

Executive Summary

This report documents a 7-weeks internship completed from June through July 2026 with the City of Oak Ridge through the University of Tennessee Institute for Public Service and its Municipal Technical Advisory Service. The internship applied quantitative economics, data analysis, financial modeling, and reporting-system design to two municipal finance assignments. The primary assignment involved the development of a sales tax data integration, automation, and reporting system for records associated with Anderson and Roane Counties. The second involved the analysis and enhancement of an energy-sector revenue projection framework used by the City to evaluate the long-term fiscal effects of major industrial and nuclear-related investments.

The sales tax assignment began with administrative records that had been collected and stored but were not part of a regular analytical process. Filing records extended back to 2009, while detailed collection records covered the most recent twelve-month period. The source files were provided primarily in Excel and OpenDocument Spreadsheet formats and contained information on company identity, registration, addresses, filing month, collection month, payment amount, filing frequency, county, business category, location, and miscellaneous tax activity. Their analytical use was limited by inconsistent formatting, missing information, timing differences, ambiguous classifications, outdated records, and structural variation across files.

Initial exploration and cleaning were performed in R. However, because the permanent system needed to remain accessible and maintainable for future municipal employees, the operational workflow was reconstructed using Excel, Power Query, SharePoint, Power BI, and DAX. Two complementary reporting environments were created. The Excel and Power Query system provided a familiar spreadsheet-based option, while the Power BI system offered more efficient processing, interactive filtering, stronger visual presentation, data-quality monitoring, and greater scalability.

The final Power BI system integrated Anderson and Roane County records into one model while preserving county-specific filters. Its report pages included a Revenue Overview, Company Revenue Share analysis, Collection versus Filing Month matrix, Company Payment Matrix, Company Filing Regularity analysis, Data Quality Review, and a six-month revenue forecast. These tools enabled users to examine total revenue, monthly patterns, geographic distribution, taxpayer concentration, filing and collection timing, missing periods, questionable records, and other matters requiring further investigation.

The system revealed several categories of anomaly, including companies registered at unexpected locations, records associated with businesses that may no longer operate under the reported name, online sellers without a physical Oak Ridge location, irregular payment timing, missing filing records, substantial miscellaneous revenue, and apparent inconsistencies between business classification and observed payment levels. These findings were treated as matters for additional verification rather than evidence of noncompliance.

A recurring monthly update can ordinarily be completed by placing a correctly structured file in the appropriate county SharePoint folder and refreshing the Power BI dashboard. Detailed maintenance instructions were developed to explain file naming, worksheet and table requirements, validation procedures, common failure points, and troubleshooting methods. The principal contribution was therefore not a one-time analysis, but a durable institutional

capability through which the City can continue using and improving its sales tax information.

The energy-sector assignment involved gathering and harmonizing information from company websites, news reports, government publications, U.S. Department of Energy materials, property viewers, surveys, municipal sources, Davenport personnel, and discussions with city and county leadership. Because access to parts of the live projection model was restricted, a separate duplicate framework was constructed using simulated assumptions. This controlled environment allowed the model's internal relationships to be examined before approved information was entered into the permitted portions of the live system.

A separate analytical workbook was also developed using linked formulas, XLOOKUP functions, PivotTables, and charts to organize annual and cumulative city and county revenue projections through 2100. Exact company names, investment assumptions, and financial results remain confidential and are not disclosed in this report. The discussion instead emphasizes the methodology, model structure, information requirements, analytical reasoning, and continuing need to update projections as investment plans, construction schedules, abatements, and tax decisions change.

The internship demonstrated that local governments may possess valuable administrative information without having the systems necessary to transform it into timely and actionable knowledge. Both assignments strengthened the City's capacity to organize data, identify information gaps, interpret fiscal patterns, evaluate future investment, and communicate complex results to decision makers. The experience also showed that effective analytical systems must be accessible, replicable, maintainable, scalable, secure, and compatible with the institution in which they will be used.

Professionally, the internship transformed my understanding of local government and strengthened my commitment to applied public policy. As an international student with limited previous exposure to the United States local-government system, I was given meaningful access to municipal operations, senior leadership discussions, public safety departments, administrative processes, and a City Council meeting. The trust provided by IPS, MTAS, and the City of Oak Ridge gave me both the confidence and the practical direction to continue applying quantitative economics and computational methods in service of public institutions and communities.

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Introduction

1.1 Background and Internship Context

Local governments routinely generate, receive, and preserve substantial quantities of administrative information through taxation, financial reporting, utilities, permitting, economic development, public safety, infrastructure management, and other municipal responsibilities. The existence of these records, however, does not guarantee that they can be used effectively. Data may remain separated across files, departments, and reporting periods, follow inconsistent structures, contain incomplete or ambiguous fields, or require significant manual preparation before they can support meaningful analysis. An important component of modern public administration is therefore not merely the acquisition of data, but the development of institutional processes through which those data can be validated, updated, interpreted, and translated into timely decisions.

This report documents a 7-weeks summer internship completed from June through July 2026 through the University of Tennessee Institute for Public Service and its Municipal Technical Advisory Service. The internship was hosted by the City of Oak Ridge, Tennessee, under the primary supervision of Deputy City Manager Tom Pessemier. Guidance and professional support were also provided by Kristen Davis, Margaret Norris, Tomi Rogers, Valerie Dugan, and other personnel associated with IPS, MTAS, and the City of Oak Ridge.

The internship centered on two applied municipal finance assignments. The first involved the creation of a repeatable sales tax data integration, analysis, automation, and reporting system for records associated with Anderson and Roane Counties. The second involved the examination and enhancement of an energy-sector revenue projection framework used to estimate the long-term municipal fiscal effects of major industrial and nuclear-related investments.

Although the assignments differed in their data sources, technical environments, operational purposes, and confidentiality requirements, they addressed a common administrative challenge. In both cases, valuable information existed, but its usefulness depended on the creation of a reliable structure through which officials could organize inputs, identify gaps, review assumptions, interpret outputs, and maintain the analysis as new information became available. The internship therefore focused not only on producing individual calculations or visualizations, but on strengthening the City's continuing capacity to use data in fiscal planning, administrative review, and policy discussion.

1.2 The Institute for Public Service and Municipal Technical Advisory Service

The University of Tennessee Institute for Public Service, commonly referred to as IPS, is the public-service outreach organization of the University of Tennessee System. Established in 1971, IPS serves governments, businesses, industries, and communities throughout Tennessee through technical assistance, professional consultation, training, and applied research. Its statewide structure connects specialized knowledge and university resources with the operational needs of public institutions and organizations serving communities across Tennessee (University of Tennessee Institute for Public Service, 2026a).

The Municipal Technical Advisory Service, or MTAS, is one of the principal IPS agencies serving local government. MTAS works directly with Tennessee municipalities and their officials in areas including municipal management, finance and accounting, legal assistance, human resources, public safety, utilities, public works, training, and organizational development. Its work is intended to strengthen municipal administration and improve the quality of services provided to Tennessee communities (University of Tennessee Institute for Public Service, 2026b).

The IPS internship program places students with local governments and public-service organizations across Tennessee to complete substantive assignments, participate in professional discussions, and gain direct exposure to the responsibilities of elected officials, appointed administrators, consultants, and municipal employees. The program is statewide rather than limited to a single city or region, allowing students to learn from the varied administrative needs of Tennessee cities, counties, and localities. It also creates opportunities for individuals from backgrounds that have historically had limited representation or access within public service (University of Tennessee Institute for Public Service, 2026c).

For an international student with rigorous theoretical and computational training in quantitative economics but limited previous exposure to the structure of local government in the United States, this arrangement provided an especially meaningful pathway into applied public policy. Entering the public sector from an international background can involve uncertainty regarding institutional expectations, governance structures, professional trust, and the practical relationship between economic analysis and public administration. IPS and MTAS created a platform through which these barriers could be addressed through substantive work rather than observation alone.

The City of Oak Ridge strengthened that opportunity by providing access to consequential municipal assignments, professional resources, leadership discussions, operational environments, and technical guidance. The internship was not restricted to predetermined clerical tasks. Instead, it involved freedom to investigate problems, test alternative technical approaches, create new analytical structures, and communicate findings to senior municipal and county officials. This combination of trust, accountability, and guidance made it possible to connect quantitative training with practical public-sector needs.

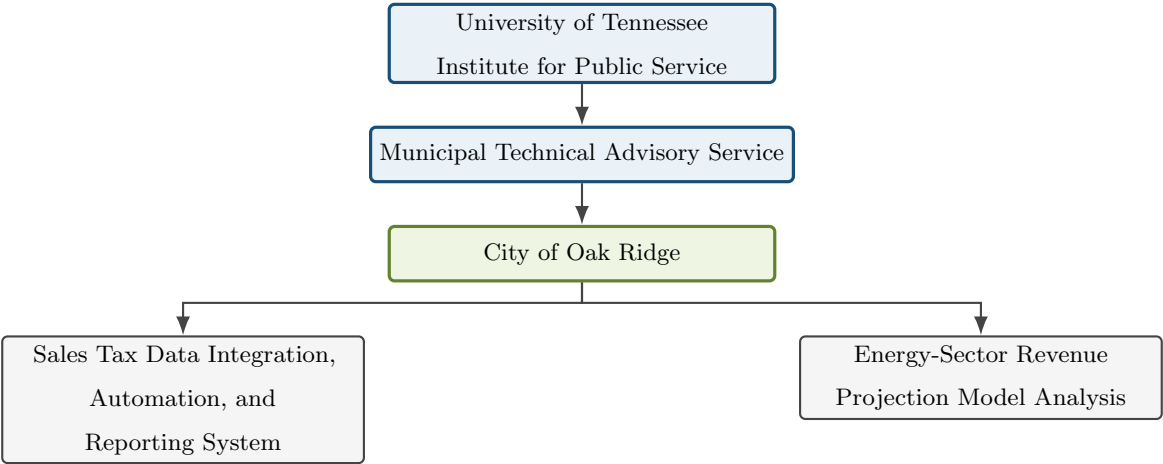


Figure 1.1: Institutional structure and principal internship assignments.

1.3 The City of Oak Ridge

Oak Ridge occupies a distinctive place in the administrative, scientific, and economic history of Tennessee and the United States. The community was established by the federal government during World War II as a central location of the Manhattan Project. Its creation brought together scientific research, industrial production, national security, infrastructure development, and a rapidly constructed residential community. Following the war, Oak Ridge transitioned from a restricted federal settlement into an independent municipality and became self-governing in 1959 under a modified council-manager form of government (City of Oak Ridge, 2026a).

The City's historical development produced an unusual relationship among municipal government, federal facilities, scientific institutions, energy policy, industrial investment, and regional economic development. That relationship continues to shape Oak Ridge's contemporary identity and administrative responsibilities. The community maintains close connections with major federal and nuclear facilities, including Oak Ridge National Laboratory, the Y-12 National Security Complex, and the former K-25 site, now known as the East Tennessee Technology Park.

Oak Ridge therefore functions not only as a residential municipality, but also as an important center for scientific research, nuclear security, environmental management, advanced manufacturing, infrastructure development, energy innovation, and emerging private-sector investment. Its municipal government must provide everyday public services while also responding to the fiscal and operational consequences of unusually large federal, industrial, technological, and economic-development activities.

The City supports this community through a broad administrative structure that includes city management, finance, fire and police protection, planning and development, public works, electric utility operations, recreation and parks, library services, communications, information systems, and other public functions (City of Oak Ridge, 2026b). During the internship, exposure to these departments and to the officials and employees responsible for maintaining them demonstrated that local government is considerably broader and more operationally complex than it may appear from outside the institution.



Figure 1.2: Oak Ridge Municipal Building, the primary location of the summer internship.

The City's effectiveness depends on continuous coordination among employees whose work is often not visible unless a problem occurs. Finance personnel maintain the fiscal systems needed to support public services. Police officers, firefighters, dispatchers, and emergency personnel remain prepared to respond at all hours. Public works employees maintain infrastructure and essential operations. Planning, development, administration, and utility personnel balance present service demands with future growth. One of the most significant lessons of the internship was that municipal stability is produced through the combined efforts of individuals pursuing highly specialized responsibilities while remaining connected to a broader public mission.

Oak Ridge's continued relationship with the nuclear sector also increases the importance of long-term fiscal preparation. On July 28, 2026, the U.S. Department of Energy identified Tennessee as one of five initial contender states for a proposed Nuclear Lifecycle Innovation Campus. Tennessee was selected alongside Utah, Oklahoma, Louisiana, and Idaho following the review of 28 applications from 26 states. The proposed campuses are intended to support activities across the nuclear fuel lifecycle, potentially including fuel fabrication, enrichment, used-fuel reprocessing and disposition, advanced reactor deployment, power generation, advanced manufacturing, and co-located data centers (U.S. Department of Energy, 2026).

The federal announcement concerned Tennessee as a state and did not designate Oak Ridge as a final host location. Nevertheless, Tennessee's selection illustrates the scale of nuclear-sector investment and industrial development that may influence communities with established nuclear infrastructure, institutions, expertise, and supply chains. The proposed campuses have the potential to attract substantial capital investment, create thousands of jobs, and generate significant state and local tax revenue. These possibilities reinforce the importance of accurate municipal

revenue forecasting, investment analysis, economic-development planning, and long-term fiscal preparation.

For Oak Ridge, major industrial and nuclear investments can involve phased construction, equipment installation, land and building improvements, depreciation schedules, assessment ratios, tax abatements, Payment in Lieu of Taxes agreements, and revenue effects distributed between city and county governments. At the same time, recurring revenue sources such as local sales taxes remain essential to understanding current fiscal conditions. The two internship assignments were therefore connected to both the City's immediate administrative needs and its longer-term position within a changing regional and national economy.

1.4 Purpose and Objectives of the Internship

The overall purpose of the internship was to apply quantitative, computational, and economic reasoning to practical municipal problems while developing a deeper understanding of local-government operations. The assignments provided hands-on experience with administrative records, municipal finance, economic-development information, analytical modeling, data transformation, reporting-system design, process documentation, and communication with nontechnical decision makers.

The sales tax assignment called for the development of a repeatable and user-friendly system capable of transforming inconsistent sales tax records into a structured format suitable for monthly analysis, comparison, monitoring, forecasting, and decision-making. The intended work included data cleaning, record alignment, historical organization, recurring updates, reporting-tool development, and process documentation.

The completed work expanded this objective into two complementary analytical systems. The first used Excel and Power Query to provide an environment familiar to employees accustomed to traditional spreadsheet-based reporting. The second used Power BI to provide faster processing, stronger visual presentation, interactive filtering, quality review, more efficient navigation, and greater scalability as additional months and records were added.

The energy-sector assignment involved the use of the Davenport revenue projection model to examine anticipated municipal and county revenues associated with energy companies locating or expanding in Oak Ridge. The work included reviewing the model's structure, gathering information from multiple external and internal sources, organizing project assumptions, examining model outputs, improving usability, and communicating long-term financial relationships.

Because portions of the live Davenport model contained protected formulas, restricted inputs, and limited editable areas, the work also required the independent reconstruction of its logical structure in a separate environment using simulated information. This duplicate framework made it possible to test relationships, understand the effects of assumptions, and identify the types of inputs required before working within the permitted areas of the live model.

Across both assignments, five practical design principles guided the work. The resulting systems needed to remain accessible to users with different levels of technical experience, compatible with the City's existing Microsoft and SharePoint environment, replicable without dependence on undocumented individual knowledge, scalable as new information became available, and sufficiently interactive to allow users to explore the data without fearing accidental or irreversible changes.

These principles influenced decisions concerning software selection, file organization, data transformation, model design, visual presentation, documentation, and handover procedures. Technical sophistication was not treated as an

objective by itself. A solution was considered successful only if it could provide useful information, operate within institutional constraints, remain understandable to future users, and continue functioning after the internship ended.

1.5 Overview of the Sales Tax Data Integration and Reporting System

The Sales Tax Data Integration and Reporting System constituted the primary focus of the internship and the largest component of this report. At the beginning of the assignment, historical sales tax records existed in Excel and OpenDocument Spreadsheet files collected from state and local administrative sources. Filing information extended back to 2009, while detailed collection information was available for the most recent twelve-month period.

The records contained potentially valuable information including company names, registration identifiers, manufacturing addresses, corporate addresses, filing months, collection months, taxable payment amounts, payment frequency, tax filed, county, business category, location information, and miscellaneous tax fields. These records were not exceptionally large in comparison with some research datasets, but they were complex because they had not been prepared consistently for analytical use.

The files contained messy and inconsistent formatting, missing information, timing irregularities, ambiguous categories, incorrect or outdated records, and structural differences across sources. Filing months and collection months did not always correspond because some payments were processed immediately while others followed monthly or delayed filing schedules. Company names and registration details also varied, and some records appeared under miscellaneous categories with limited descriptive information.

Before the internship, these files were primarily stored rather than regularly analyzed. There was no established reporting system through which staff could efficiently combine the records, examine historical patterns, identify missing information, compare Anderson and Roane Counties, review company behavior, or generate visual summaries. The challenge was therefore not simply to improve an inefficient reporting process. It was to create an analytical process where none had previously existed.

The initial approach used R to explore, clean, filter, and combine portions of the data. R was effective for understanding the records and testing potential transformations. However, it was not selected as the permanent operational platform because future users might not have the same programming background, and a code-dependent workflow could reduce accessibility and continuity after the internship.

The system was therefore reconstructed through Excel and Power Query and later extended into Power BI. Anderson and Roane County records were integrated into one analytical environment while retaining county filters. Users could examine the combined municipal picture, compare the two counties directly, or isolate one county according to their needs.

The final reporting environment included revenue summaries, monthly collection trends, taxpayer-location categories, company revenue shares, filing and collection comparisons, payment matrices, filing-regularity indicators, company-name review flags, data-quality controls, and a six-month revenue forecast. Rather than presenting isolated numbers, the system allowed users to move quickly from a broad summary to a more detailed investigation.

The automation was designed so that a future monthly update would ordinarily require only the placement of a correctly structured file into the appropriate SharePoint folder followed by a Power BI refresh. When a file arrived under a different name or format, the maintenance instructions explained how to rename it, place the data into the

required Excel structure, and complete the update.

Detailed documentation was prepared to explain file naming, worksheet and table requirements, upload procedures, validation checks, likely failure points, and troubleshooting methods. The objective was not merely to produce a one-time analysis, but to create a sustainable municipal capability through which future staff could continue identifying trends, missing records, irregular filing behavior, questionable classifications, and areas requiring further investigation.

1.6 Overview of the Energy-Sector Revenue Projection Assignment

The Davenport Energy-Sector Revenue Projection assignment formed the second major component of the internship. The model was designed to estimate the long-term city and county fiscal effects of significant energy-sector developments. Relevant assumptions included project timelines, equipment values, installation years, useful equipment life, depreciation, land values, building improvements, assessment ratios, tax rates, abatements, and other financial conditions.

Information was gathered from company websites, news reports, government documents, U.S. Department of Energy materials, property viewers, surveys, municipal sources, discussions with Davenport personnel, and guidance from city and county officials. This information was not always available in a single document or standardized form. It had to be compared, reconciled, interpreted, and organized before it could be used effectively.

The production model included access restrictions because it was designed for use across government offices and contained formulas, structures, and information that were not intended for unrestricted modification. Rather than experimenting directly within the protected system, a separate duplicate framework was constructed using simulated assumptions.

This replicated environment functioned as a controlled analytical workspace. It allowed the relationships among investment values, installation schedules, depreciation, abatements, assessed value, tax rates, and annual revenue to be examined without creating risk for the live model. Through this process, the model's logic could be understood and partially reverse-engineered before approved information was entered into the permitted sections of the production system.

After developing this understanding, relevant information was transferred into the live model. A separate analytical workbook was also developed using linked formulas, XLOOKUP functions, PivotTables, and charts. This workbook consolidated outputs and allowed annual and cumulative city, county, company, and combined tax projections to be examined through the year 2100.

Exact company assumptions, investment values, and projected financial outcomes remain confidential and are not reproduced in this report. The discussion instead focuses on the methodology, model architecture, information requirements, analytical reasoning, data limitations, and importance of maintaining an updateable fiscal-planning framework as investments, abatements, construction schedules, and tax decisions change.

The original assignment instructions included scenario analysis. After discussion with the primary supervisor, formal scenario analysis was removed from the required project scope because it was not considered necessary for the final deliverables. The completed work instead concentrated on information collection, model understanding, model reconstruction, input development, output organization, and the creation of a framework that could be updated as

more reliable information became available.

1.7 Collaboration and Institutional Learning

Both assignments were iterative rather than linear. Their development required repeated consultation with the Deputy City Manager, the City Manager's Office, county representatives, IPS and MTAS personnel, and other municipal officials. These discussions clarified institutional needs, refined assumptions, identified confidentiality boundaries, improved the presentation of results, and ensured that the final products addressed operational rather than purely technical concerns.

Meetings with the City Manager, Deputy City Manager, county representatives, and other senior officials were especially important to the energy-sector assignment. The significance of the model could not be understood solely from its formulas. Leadership guidance explained how investment assumptions, tax policies, abatements, infrastructure needs, and development timelines could influence actual government planning.

The freedom provided by supervisors was equally important. The internship did not begin with a complete technical roadmap. There was no predetermined method for transforming the raw sales tax files into an automated system, and the constraints of the SharePoint environment created repeated technical difficulties. The work therefore required experimentation, adaptation, and professional judgment.

SharePoint was retained because it reflected the City's existing preferences concerning security, shared access, file control, and continuity. At the same time, it had less processing capacity than the available personal and office computers and sometimes experienced connectivity, cloud-editing, and refresh problems. These limitations interrupted Power Query operations and occasionally required processes to be restarted.

The restrictions on using confidential files outside approved devices and locations also meant that troubleshooting had to occur within the City's operational environment. After completing the Power Query system, the Power BI system was developed with supervisory approval because the volume of information was expected to continue growing, and a spreadsheet-centered SharePoint process could become increasingly inefficient.

This experience demonstrated that public-sector technical work cannot be separated from institutional realities. Security, accessibility, compatibility, continuity, staff capacity, and existing organizational practices must be considered alongside speed and analytical power. The most technically advanced solution is not necessarily the most useful if it cannot be maintained, trusted, or incorporated into regular operations.

1.8 Significance of the Internship

The primary contribution of the internship was not the production of a single analysis. It was the creation of durable analytical capacity. The City already possessed valuable information, but without an organized system, much of its practical usefulness remained inaccessible.

The sales tax system made it possible to identify trends, compare counties, distinguish filing and collection timing, review company payment behavior, examine location categories, forecast future revenue, and detect records requiring additional investigation. It also helped reveal what information was missing and what further data would be needed to address more complex questions.

The energy-sector work created a structured approach for understanding long-term investment and tax rela-

tionships within a restricted municipal forecasting environment. The duplicate model, linked analysis, organized assumptions, and documented methodology created a foundation that could be updated as projects, investments, abatements, and policy decisions evolved.

These capabilities are relevant beyond Oak Ridge. Municipalities across Tennessee often receive administrative and sales tax information in formats that may not be immediately suitable for analysis. A general workflow based on standardization, accessible tools, recurring updates, interactive reporting, quality control, and clear documentation could be adapted by other local governments facing similar challenges.

The practical importance of this capacity extends beyond convenience or staff time. Municipalities compete for business investment, infrastructure opportunities, economic development, and long-term growth. A government that can organize its information, understand its revenue position, identify unusual patterns, communicate fiscal reliability, and evaluate future opportunities may be better positioned to make timely decisions and engage potential investors.

In this sense, data capacity is also policy capacity. When useful information cannot be accessed or interpreted, a city may lose opportunities, insight, and institutional knowledge. In a competitive development environment, one jurisdiction's missed opportunity may become another jurisdiction's gain. Creating systems that allow information to be used consistently therefore contributes to both administrative efficiency and the broader strategic position of the community.

1.9 Organization of the Report

The remainder of this report is organized into five chapters. Chapter 2 establishes the institutional and policy context for municipal data use, sales tax analysis, and long-term revenue planning. It discusses administrative data as a public asset, municipal fiscal decision-making, Tennessee sales tax structures, economic-development incentives, and the relationship between nuclear investment and local-government planning.

Chapter 3 presents the Sales Tax Data Integration and Reporting System. It examines the original condition of the files, the available variables, the initial use of R, the selection of institutionally appropriate tools, data cleaning and standardization, the integration of Anderson and Roane County records, Power Query development, SharePoint constraints, Power BI architecture, DAX calculations, dashboard pages, quality controls, anomalies, forecasting, automation, documentation, and recommendations.

Chapter 4 examines the Davenport Energy-Sector Revenue Projection assignment. It discusses information collection, source validation, confidentiality and access restrictions, reconstruction of the model using simulated assumptions, interpretation of the model's internal relationships, application within the live environment, linked analysis using XLOOKUP and PivotTables, long-term projections through 2100, collaboration with municipal and county leadership, limitations, and future updating.

Chapter 5 presents the professional development and reflection arising from the internship while chapter 6 concludes the report by evaluating the continuing value of the systems created for Oak Ridge and their broader relevance to municipalities across Tennessee.

Institutional and Policy Context

2.1 Municipal Data as an Administrative Asset

Municipal governments depend on information generated through routine administrative activity. Tax records, utility accounts, business registrations, property information, permits, service requests, financial transactions, and economic-development documents are initially created for operational or legal purposes, but they can also support planning and policy analysis. When these records are consistently organized, local officials can use them to monitor revenue, identify changes in economic activity, compare geographic areas, detect irregularities, and evaluate the possible fiscal effects of future development.

The policy value of administrative data depends on more than its availability. Records collected by different organizations or at different times may follow incompatible structures, use inconsistent terminology, contain outdated entries, or record the same activity under different dates and classifications. These problems can prevent staff from obtaining reliable answers even when the underlying information is technically available. Municipal data capacity therefore includes the ability to clean and standardize records, preserve their history, document their meaning, connect related sources, and develop tools through which employees can retrieve useful information without rebuilding the analysis each time a question arises.

This distinction is especially important in local government because municipal employees often work under substantial time and resource constraints. A dataset that requires specialized programming, extensive manual preparation, or knowledge held by only one employee may have limited institutional usefulness. A well-designed system should instead preserve the complexity needed for accurate analysis while presenting results in a form that can be understood and maintained by future users. Accessibility, documentation, and continuity are therefore components of analytical quality rather than secondary conveniences.

The sales tax assignment addressed this institutional gap directly. The City possessed historical records containing valuable information about taxpayers, filing periods, collection periods, business locations, payment amounts, and business classifications. However, the information was distributed across inconsistent files and was not connected to a regular analytical process. The purpose of the project was consequently not simply to summarize existing records, but to convert stored administrative information into a continuing municipal resource.

2.2 Sales Tax and Municipal Fiscal Decision-Making

Sales and use taxes represent an important source of revenue for Tennessee local governments. Tennessee applies a general state sales tax rate of 7 percent to most taxable tangible personal property and services, while local jurisdictions may levy additional local sales and use taxes up to a maximum rate of 2.75 percent. Filing schedules may be monthly, quarterly, or annual depending on the taxpayer's required frequency, which can create differences in the timing and pattern of reported collections (Tennessee Department of Revenue, 2026).

Under Tennessee's local option sales tax structure, the state Department of Revenue collects local sales taxes and distributes the proceeds through the applicable county and municipal framework. When a county levies the tax,

one-half of the distributed local proceeds is allocated for school purposes, while the other half is generally distributed according to the jurisdiction in which the taxable sale occurred, unless an alternative distribution arrangement has been established. Collections associated with incorporated municipalities therefore have direct implications for municipal general-fund revenue (Municipal Technical Advisory Service, 2025b).

This collection and distribution system creates several analytical challenges for local governments. Municipal staff may receive records after the underlying business activity occurred, and the month in which a taxpayer files may differ from the month in which the related revenue is collected or reported to the City. Taxpayers also operate under different filing frequencies. A company filing monthly will produce a different record pattern from one filing quarterly or annually, even when both remain compliant with their respective requirements.

The distinction between filing month and collection month is therefore important. Collection-month analysis supports the monitoring of revenue received during a municipal reporting period, while filing-month analysis helps identify the period to which a business return relates. Treating the two fields as interchangeable could distort trends, particularly when payments are delayed, adjusted, reported irregularly, or associated with prior periods. Preserving both fields in the analytical system allows officials to examine current cash-flow patterns while also investigating the timing of underlying taxpayer activity.

Geographic classification creates another layer of complexity. Oak Ridge extends into both Anderson and Roane Counties, requiring records from the two counties to be preserved and compared within a common framework. Businesses may also be associated with Oak Ridge, another Tennessee jurisdiction, an out-of-state address, or an unclear location. Online commerce further complicates interpretation because tax revenue may be connected to sellers with no physical location inside the City. A useful reporting system must therefore allow staff to examine county, taxpayer location, registration information, and payment history together rather than interpreting any one field in isolation.

Sales tax information can support more than routine revenue reporting. Changes in monthly collections may indicate shifts in consumer activity, business performance, seasonality, or reporting timing. Company-level analysis can reveal concentration among major taxpayers and help officials understand whether municipal revenue depends heavily on a relatively small group of businesses. Filing-regularity analysis can identify missing periods or unusual payment behavior, while comparisons between business classifications and observed payments can identify records requiring further verification.

These analytical uses do not establish noncompliance or wrongdoing by themselves. Administrative records may contain delayed filings, amended returns, classification errors, business closures, registration changes, or other legitimate explanations. The appropriate purpose of the system is therefore to identify patterns requiring review rather than to draw unsupported conclusions. The dashboard developed during the internship was designed around this principle by distinguishing routine reporting from data-quality flags and investigative questions.

2.3 Local Sales Tax Collection and Data Timing

Tennessee law requires the Department of Revenue to collect local sales tax together with the state sales tax. The state retains an authorized portion for administrative expenses before forwarding the remaining local proceeds through the county distribution process (Municipal Technical Advisory Service, 2025a). As a result, municipalities

do not directly control every stage through which taxpayer information and payments move before appearing in local records.

This institutional separation helps explain why municipal sales tax files may contain timing differences, incomplete descriptions, unfamiliar business names, or transactions associated with businesses outside the immediate jurisdiction. The files reflect a broader tax-administration process rather than a locally designed database created specifically for municipal analysis. Consequently, the structure that is adequate for state collection or clerical distribution may not be sufficient for local trend analysis and decision support.

Different filing frequencies also produce expected gaps in company-level monthly records. A missing payment in one month does not automatically represent a problem if the taxpayer files quarterly or annually. However, when payment frequency, filing history, and expected reporting periods are considered together, staff can distinguish potentially normal gaps from records that may warrant further review. This was one reason the project preserved payment frequency and constructed company filing-regularity measures rather than treating every absent month in the same manner.

The system also preserved negative amounts, adjustments, miscellaneous entries, and small online payments rather than automatically removing records that appeared unusual. Negative values may represent corrections or reversals, while small amounts can reflect legitimate online transactions. Retaining these records protects the integrity of the source information and allows users to separate company revenue from adjustments through report filters. Data cleaning was therefore intended to standardize and clarify the records without altering their underlying financial meaning.

2.4 Municipal Reporting, Forecasting, and Decision Support

Traditional financial reporting generally answers what has already occurred. A decision-support system extends that function by allowing users to explore why a pattern may have occurred, where it was concentrated, whether it appears unusual, and what may happen next. The distinction is important because municipal officials often need to move quickly between summary information and detailed records during budget preparation, leadership discussions, economic-development meetings, or questions concerning a particular taxpayer or period.

The reporting environment developed during the internship was designed to support both executive review and deeper analysis. A senior official could use the Revenue Overview page to identify total collections, county distribution, monthly trends, geographic sources, revenue type, and the six-month forecast within a single view. Staff requiring greater detail could move to pages examining company revenue shares, collection and filing months, payment patterns, filing regularity, and data-quality concerns.

Interactive filters increase the practical value of these pages. Users can isolate Anderson County, Roane County, a selected collection period, or a particular revenue type without changing the underlying dataset. This allows the same reporting system to answer multiple questions and reduces the need to create separate spreadsheets for every meeting or request. It also provides a safer environment for exploration because ordinary filtering does not overwrite the source records or formulas.

Forecasting adds a prospective component to the system, but projected values must be interpreted cautiously. A short historical collection period, irregular payments by major taxpayers, delayed filings, adjustments, and unusual

economic events can influence a forecast. The purpose of the six-month projection is therefore not to guarantee future revenue, but to provide a transparent baseline based on the pattern currently available. As additional monthly records accumulate, the forecasting function can be reassessed and improved.

2.5 Economic Development Incentives and Long-Term Fiscal Planning

Municipal fiscal analysis also extends beyond recurring revenues to the effects of major development projects. Large industrial and energy-sector investments may generate property tax revenue, employment, infrastructure demand, utility activity, and secondary economic effects over many decades. Their fiscal consequences depend on the timing and value of land acquisition, construction, building improvements, equipment installation, depreciation, assessment ratios, tax rates, and negotiated incentives.

In Tennessee, industrial development boards may support economic development through tools including Payments in Lieu of Taxes and Tax Increment Financing. These mechanisms can reduce or redirect portions of the tax burden associated with a project in exchange for anticipated employment, investment, redevelopment, or other public benefits. Because such arrangements affect the timing and amount of revenue available to local governments, they require careful documentation and long-term analysis (Tennessee Comptroller of the Treasury, 2026b).

A Payment in Lieu of Taxes arrangement generally arises when property is owned by a tax-exempt governmental or industrial development entity and leased to a private company. Rather than paying the full property tax that would otherwise apply, the company makes negotiated payments according to an approved agreement. The amount and duration of the benefit can vary according to project characteristics and local policy. For fiscal forecasting, the relevant issue is not simply the stated value of the investment, but the taxable or payment-generating value remaining after ownership structure, assessment, depreciation, and abatement assumptions are applied.

Tax Increment Financing uses anticipated growth in property tax revenue to support development or redevelopment costs. A base level of property value and associated revenue is established before the project, and some or all of the incremental revenue generated above that base may be used to finance eligible improvements. Tennessee requires state review of certain TIF plans to determine whether the financing serves the state's interest (Tennessee Comptroller of the Treasury, 2026a).

Although the internship did not require the completion of formal scenario analysis, the Davenport model still needed to represent the relationships through which project assumptions affect future city and county revenues. Equipment may be installed in phases, decline in taxable value through depreciation, or receive temporary abatements. Land and building values may enter the tax base in different years, while city and county tax rates produce separate revenue streams. A long projection horizon is necessary because some projects may operate, expand, replace equipment, or alter their incentive status over many decades.

2.6 Property Assessment and Projection Structure

Property tax projections require a distinction between appraised value and assessed value. Tennessee classifies industrial and commercial property at an assessment ratio of 40 percent of appraised value. The assessed value, rather than the full appraised value, is multiplied by the applicable property tax rate to determine the tax obligation before considering abatements or other arrangements (Tennessee Comptroller of the Treasury, 2026c).

Personal property and real property must also be modeled separately. Personal property generally includes machinery, equipment, and other business assets whose taxable value may decline over their useful life. Real property includes land, buildings, and improvements, which may follow different valuation and timing assumptions. A project involving several construction and equipment phases therefore requires a year-by-year structure capable of recognizing when each component enters the tax base and how its value changes over time.

The Davenport framework incorporates these relationships through project-level assumptions and annual calculations. Equipment values are connected to installation years, useful-life assumptions, depreciation, assessment ratios, abatements, and city and county tax rates. Land and building components are connected to acreage, value, construction phases, completion years, growth assumptions, assessment ratios, and incentive periods. The outputs then estimate annual net taxes and cumulative revenue over the projection horizon.

The use of a model does not remove uncertainty. Company plans can change, construction schedules can shift, equipment values can be revised, tax rates can change, and incentive agreements may be amended. A useful model must therefore remain updateable and clearly distinguish factual inputs from assumptions. Documentation is particularly important because a future user must be able to determine where an estimate originated, why it was selected, and which output would change if the information were revised.

2.7 Oak Ridge and Applied Fiscal Capacity

The importance of these analytical capabilities is amplified by Oak Ridge's economic position. The City must manage ordinary municipal responsibilities while operating within an environment influenced by federal facilities, nuclear-sector investment, large industrial projects, scientific institutions, and activity distributed across two counties. These conditions create opportunities for growth but also increase the complexity of revenue analysis and long-term planning.

The sales tax and energy-sector assignments address different sides of the same fiscal responsibility. The sales tax system examines recurring revenue already being reported and collected, while the Davenport framework evaluates revenue that may emerge from future investment and development. One strengthens understanding of present municipal conditions, and the other supports preparation for possible long-term change.

Together, the two assignments illustrate that municipal data systems are not merely technical products. They shape what questions officials can answer, how quickly they can respond, what information they can present to leadership, and how confidently they can evaluate new opportunities. Building local analytical capacity therefore contributes to accountability, continuity, strategic planning, and the effective stewardship of public resources.

Sales Tax Data Integration and Reporting System

3.1 Project Background and Problem Definition

The Sales Tax Data Integration and Reporting System constituted the principal technical assignment of the internship. The City of Oak Ridge possessed sales tax records associated with business activity in both Anderson and Roane Counties, but the records had not been organized into a recurring analytical process. Files were preserved as administrative data rather than maintained within a structured reporting environment. As a result, staff could not readily use the available information to examine revenue trends, compare the two counties, investigate payment timing, assess company filing behavior, identify incomplete records, or generate consistent visual summaries.

The project began from the recognition that possessing data and being able to use data are different institutional capabilities. The raw files contained information with significant administrative and fiscal value, but their structure limited their practical application. The central problem was therefore not a lack of information. It was the absence of a repeatable framework through which the information could be cleaned, connected, refreshed, interpreted, and maintained.

The original internship assignment anticipated the creation of a user-friendly tool capable of integrating state-provided sales tax information with locally maintained records. The work ultimately developed into a broader municipal reporting system. It included historical data preparation, an Excel and Power Query reporting environment, a Power BI data model, interactive dashboard pages, quality-control measures, forecasting, a SharePoint-based update process, and detailed maintenance and troubleshooting documentation.

The completed system was designed to serve several audiences simultaneously. Senior officials needed a concise overview that could communicate the City's revenue position within seconds. Finance and administrative staff needed the ability to investigate individual companies, filing patterns, collection timing, and questionable records. Future employees needed a process that could be updated without reconstructing the system or possessing the same programming background as the original developer. These requirements shaped the technical architecture and visual design of the final product.

3.2 Data Sources and Historical Coverage

The project used sales tax information obtained through state and local administrative sources. The records were provided primarily through Microsoft Excel and OpenDocument Spreadsheet files. Tax filing information extended back to 2009, while detailed tax collection information available for the reporting system covered the most recent twelve-month period from July 2025 through June 2026.

The available variables included company name, registration identifier, manufacturing address, corporate address, filing month, collection month, taxable payment amount, payment frequency, tax filed, county, business category, location information, miscellaneous descriptions, and other tax-related fields. Not every source file contained every variable in the same form, and the meaning of some fields had to be inferred through comparison with related records and consultation with municipal staff.

The records were associated with both Anderson and Roane Counties because the City of Oak Ridge extends into both jurisdictions. Anderson County accounted for a larger share of the observed commercial activity and collected revenue, which was broadly consistent with its greater commercial concentration. Roane County nevertheless included several meaningful taxpayers and revenue patterns that could not be treated as negligible. Preserving both counties within the same model was therefore necessary for a complete municipal analysis.

The combined system retained county as an analytical field instead of creating entirely separate reporting products. This allowed users to view the complete dataset, compare Anderson and Roane Counties directly, or isolate one county through filters. Maintaining a unified structure also ensured that identical transformation rules, measures, classifications, and quality-control procedures were applied to both sources.

Table 3.1 summarizes the principal categories of information incorporated into the reporting system. The precise availability and structure of individual fields varied across source files, but the final model retained the variables required for company, geographic, temporal, financial, and data-quality analysis.

Table 3.1: Principal fields incorporated into the sales tax reporting system

Field category	Examples and analytical use
Company identification	Company name, cleaned company name, registration identifier, location identifier, and source-file name. These fields support company-level matching, review, and traceability.
Location information	Business address, corporate address, manufacturing address, county, and derived location category. These fields support geographic analysis and location validation.
Time information	Filing period, filing month, collection month, month number, and year. These fields support trend analysis and the examination of timing differences.
Financial information	Revenue collected, tax filed, payment amount, adjustments, miscellaneous tax activity, and company revenue share.
Filing information	Payment or filing frequency, observed filing months, missing-month count, and payment regularity indicators.
Business classification	Business category, industry description, company type, and miscellaneous classification information.
Quality-control information	Company-name review status, unknown-location indicator, missing identifier, source record, and other fields used to identify records requiring review.

3.3 Original Condition of the Records

The primary difficulty was not the number of observations. The dataset was substantially smaller than several academic and research datasets previously handled during graduate study. Its complexity instead arose from the condition and heterogeneity of the records. Files from different periods and sources had not been prepared according to one stable analytical standard.

Company names contained inconsistent punctuation, abbreviations, legal suffixes, capitalization, spacing, and

occasional descriptive text. Some names appeared incomplete or were entered in ways that resembled dates or other non-company values. Registration information was sometimes missing, outdated, or difficult to connect across files. Business categories did not always appear consistent with the observed scale of the taxpayer, and some large payments were associated with classifications that appeared unusually small.

Address information created additional complications. Some companies appeared to be registered at locations different from those expected based on their known operations. Other records were associated with companies that no longer appeared to exist or operate under the reported name. Online businesses could remit sales tax to Oak Ridge despite having no physical location in the City. These situations required the system to preserve distinctions among Oak Ridge, out-of-state, other Tennessee, and unknown or unclear locations.

Timing fields also required careful treatment. The month in which a taxpayer filed a return was not always the same as the month in which the related revenue appeared in the collection records. Some transactions were processed quickly, while other taxpayers followed monthly, quarterly, delayed, or irregular filing patterns. Adjustments and corrections could also appear in later periods, including negative values. A reliable model therefore had to preserve filing month and collection month as separate variables.

A substantial portion of revenue was also recorded through miscellaneous business or adjustment categories for which limited descriptive information was available. These records could not simply be deleted because they represented actual financial activity. The challenge was to retain them while separating them from clearly identified company revenue so users could understand how much of the total remained difficult to classify.

The condition of the records demonstrated that data cleaning could not be limited to removing blanks or changing formats. It required the preservation of financial meaning while developing consistent names, dates, categories, geographic classifications, and relationships that could support analysis.

3.4 Initial Exploration Using R

The initial technical approach used R because it provided an efficient environment for inspecting, filtering, cleaning, and combining inconsistent files. Scripts were developed to explore column structures, standardize selected variables, compare records, and test whether external information could be accessed through application programming interfaces.

R was particularly useful during the exploratory stage because transformations could be tested rapidly and repeated across multiple files. It also helped clarify which inconsistencies could be resolved systematically and which required institutional interpretation or manual review. This stage provided a clearer understanding of the data before a permanent municipal system was selected.

Despite its analytical advantages, R was not retained as the principal operational platform. The long-term usefulness of the project depended on whether future employees could maintain the system after the internship ended. A workflow requiring programming knowledge could create dependence on one technically specialized user and reduce the likelihood that the system would be updated consistently.

The decision to move away from R as the permanent platform was therefore not a rejection of its technical capability. It reflected a distinction between a tool suitable for an individual analyst and a tool suitable for institutional adoption. R remained valuable for initial exploration, but Excel, Power Query, SharePoint, and Power

BI offered a closer fit with the City’s existing software environment and the anticipated needs of future users.

3.5 Design Principles and Tool Selection

Five design principles guided the development of the permanent system. First, the workflow needed to be accessible. Staff members with strong spreadsheet experience but limited programming exposure needed to be able to understand the process and review the outputs.

Second, the system needed to be replicable. A future employee should not have to rely on undocumented knowledge held by the original developer. File requirements, transformation logic, update procedures, and troubleshooting steps needed to be defined clearly enough for another user to reproduce the process.

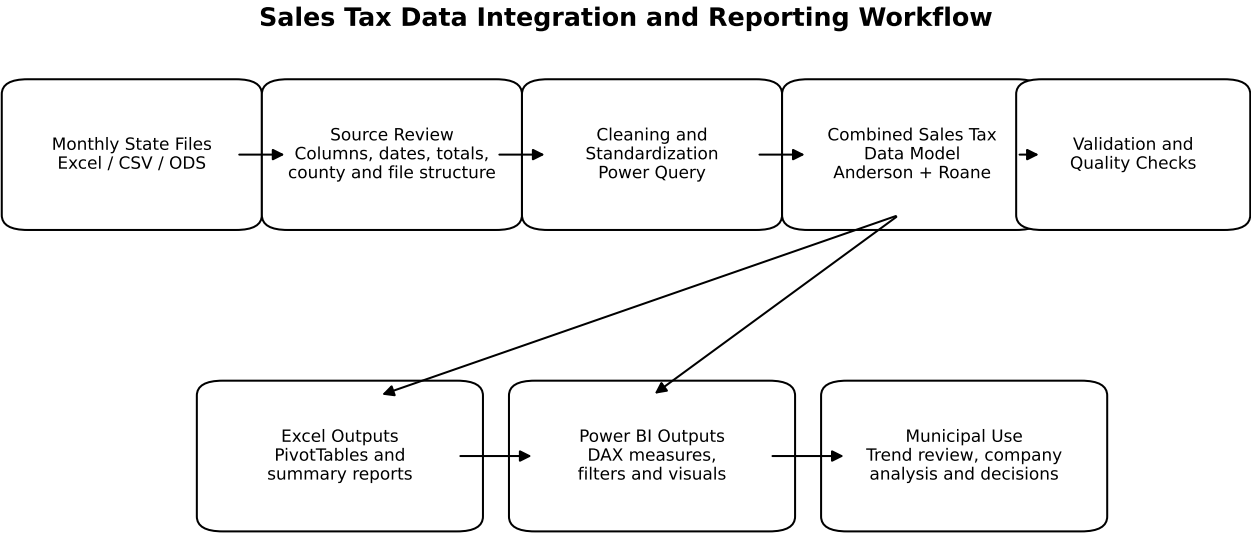
Third, the system needed to be maintainable. Municipal data systems frequently outlast the employees or interns who first create them. The solution therefore had to continue operating after the internship and include guidance for identifying and correcting common failures.

Fourth, the system needed to be scalable. The historical record would continue growing as new monthly files were received. A solution that performed adequately with one year of detailed collection records might become inefficient after several additional years. The architecture needed to anticipate future volume, new questions, additional calculations, and revised reporting needs.

Fifth, the system needed to support safe analytical exploration. Users should be able to filter counties, dates, companies, revenue types, and location categories without changing the underlying records or fearing irreversible damage. This principle was especially important for encouraging nontechnical users to engage directly with the data.

These considerations led to the development of two complementary systems. Excel and Power Query provided a familiar spreadsheet-based environment, while Power BI offered a lighter and more scalable reporting layer with stronger interactivity, visualization, and data-modeling capabilities. The two environments were not treated as competing products. They served different user preferences and operational needs.

The complete analytical workflow is summarized in Figure 3.1. The process begins with recurring administrative files and proceeds through structural review, cleaning, standardization, integration, validation, reporting, and municipal use. Although several software platforms were involved, the components were designed as one connected reporting process rather than as separate analytical products.



The workflow converts recurring administrative files into a documented, refreshable reporting system.

Figure 3.1: Sales tax data integration and reporting workflow.

3.6 Data Cleaning and Standardization

The cleaning process began with a review of the structure of each source file. Column headings, worksheet names, table structures, date formats, numeric fields, totals, and descriptive variables were examined before the records were combined. Files containing total or grand-total rows required those rows to be removed so that summary values would not be interpreted as individual transactions.

Company-name preparation required trimming leading and trailing spaces, standardizing capitalization, removing inconsistent formatting where appropriate, and creating a cleaned name field for analysis. The original source name was preserved when necessary so that users could compare the cleaned version with the source record. This protected traceability while allowing records with minor formatting differences to be grouped more consistently.

Date fields were converted into stable filing-month and collection-month structures. The system preserved both variables because they represented different stages of the tax process. Additional year and month fields were created to support chronological sorting and visualization. This was necessary because text-based month labels can sort alphabetically rather than chronologically if they are not connected to an underlying date field.

Revenue amounts were converted into numeric formats while preserving negative values, small payments, adjustments, and zero values where they reflected actual source information. Unusual values were not automatically excluded. Removing them could have altered the financial meaning of the dataset and concealed corrections, reversals, or infrequent online transactions.

County information was standardized into Anderson County and Roane County categories. Location information was also grouped into analytically useful sections, including Oak Ridge, out of state, Tennessee but not Oak Ridge,

and no location or unknown. These categories allowed staff to examine whether revenue originated from companies physically associated with the City, located elsewhere in Tennessee, operating from outside the state, or lacking sufficient location information.

Revenue type was divided between identifiable company activity and other or adjustment activity. This distinction allowed users to include all revenue in total calculations while separately examining the amount that could not be associated confidently with a standard company record.

The cleaning process was intentionally conservative. The objective was not to rewrite the source data according to assumptions that could not be verified. Instead, the system standardized fields needed for analysis, preserved unusual values, and created review indicators where records required further investigation.

3.7 Excel and Power Query Reporting System

The first permanent implementation was developed in Excel using Power Query. This environment was selected because it was already familiar to municipal staff and did not require the installation or adoption of a specialized programming platform. Power Query allowed multiple source files to be imported, transformed, appended, and refreshed through a repeatable sequence.

Separate county source folders were established for Anderson and Roane County files. Power Query read the files within each folder, applied standardized transformations, and combined the records into a broader analytical dataset. The transformation sequence included column selection, data-type conversion, removal of summary rows, date preparation, company-name cleaning, location classification, revenue-type assignment, and county labeling.

PivotTables were then used to summarize revenue by county, month, company, location category, and other dimensions. PivotCharts provided an Excel-based visual reporting layer for users who preferred to remain within the traditional spreadsheet environment.

The Excel and Power Query reporting environment is illustrated in Figure 3.2. It provided a familiar spreadsheet-based interface through which users could review county, company, location, and monthly revenue summaries.

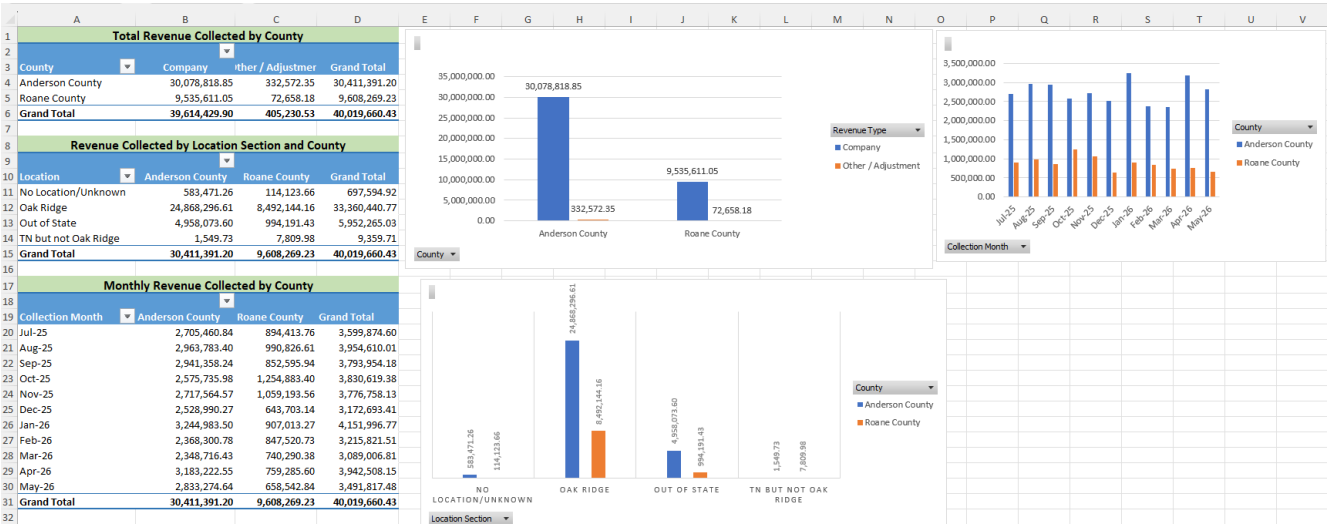


Figure 3.2: Excel and Power Query reporting environment showing county, location, and monthly revenue summaries.

The Excel system demonstrated that the source files could be converted into recurring reports without manually

rebuilding every summary. Users could refresh the queries after new information was added and interact with PivotTable filters to examine selected periods, counties, locations, or revenue types.

The Excel version nevertheless revealed several long-term limitations. As the number of files and calculations increased, refresh operations became more demanding, particularly when executed through the City's SharePoint environment. Large PivotTables also required substantial worksheet space and could become difficult to navigate when filing records extended back to 2009.

These limitations did not make the Excel system unsuccessful. It remained a useful and familiar option. They did, however, demonstrate the value of developing an additional reporting environment capable of handling the growing data load more efficiently.

3.8 SharePoint Architecture and Operational Constraints

The system needed to operate through SharePoint because it was the City's preferred environment for centralized access, document control, continuity, and security. This institutional requirement was appropriate given the restricted and administrative nature of some of the records. It also ensured that the completed work would remain available to authorized staff after the internship.

SharePoint provided several important benefits. It centralized the county source files, allowed the reporting products to be stored in a shared location, supported controlled access, and reduced dependence on a single employee's local computer. It also created a clear folder structure through which future monthly records could be incorporated into the system.

At the same time, SharePoint created the most difficult technical challenge in the sales tax assignment. The available personal and office computers possessed substantially greater processing capacity than the cloud-based directory. Complex Power Query operations could be interrupted by connectivity problems, cloud editing limitations, or refresh failures. In several instances, work had to be restarted after a query or command failed before completion.

Security restrictions also limited the ability to transfer confidential records to personal devices or continue troubleshooting outside the approved work environment. These restrictions were necessary, but they increased the time required to test and rebuild processes. The development of the Power Query system consequently required several weeks of repeated adjustment and validation.

The SharePoint experience influenced the decision to develop the Power BI system after the Excel environment had already been completed. As the volume of historical information increased, relying exclusively on spreadsheet-based refreshes through SharePoint could reduce efficiency. Power BI provided a more appropriate analytical layer while allowing SharePoint to remain the secure source repository.

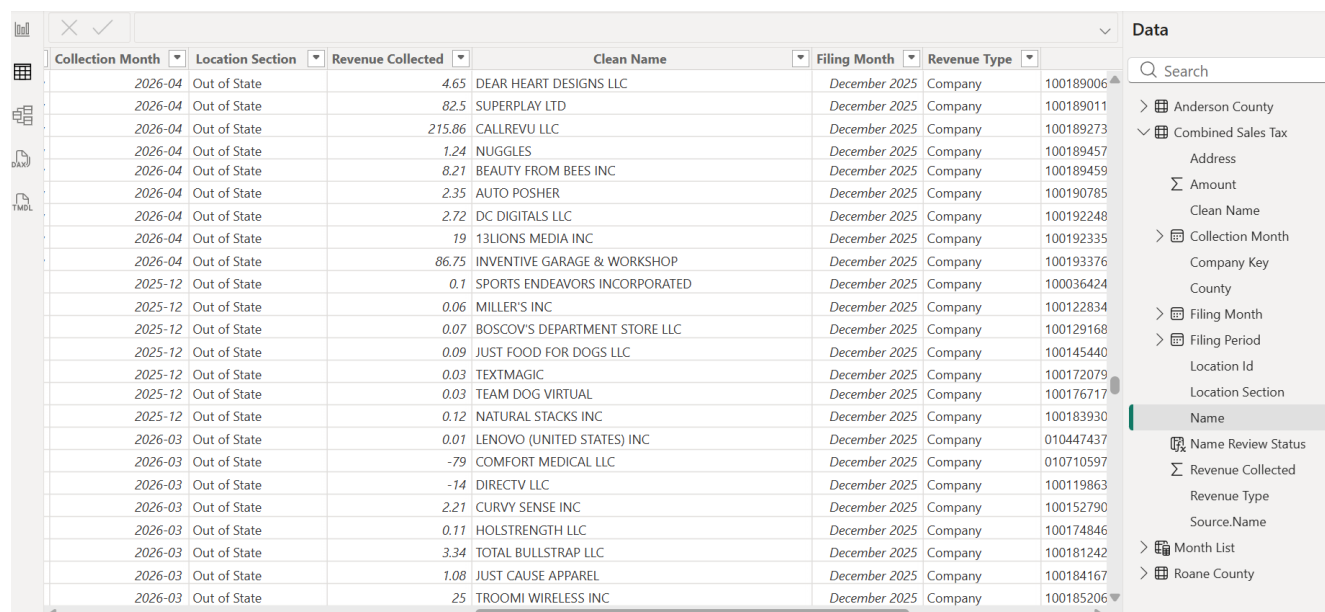
3.9 Power BI Data Model Development

The Power BI system was created from scratch after consultation with the primary supervisor. No existing municipal Power BI dashboard or data model was available for the project. The development process therefore required the creation of source connections, query transformations, table relationships, calculated columns, measures, visual pages, filters, and navigation.

The model imported Anderson and Roane County records from their respective SharePoint folders. Source

names and county identifiers were retained so that records could be traced to their originating files. The county datasets were then appended into a combined sales tax table containing the fields required for reporting.

The model preserved the separate county source tables while using the combined table for most calculations and visuals. A month-list structure was also used to support the identification of missing filing periods and chronological comparisons. This architecture allowed historical filing records and current collection records to be examined through a common reporting environment.



Collection Month	Location Section	Revenue Collected	Clean Name	Filing Month	Revenue Type	
2026-04	Out of State	4.65	DEAR HEART DESIGNS LLC	December 2025	Company	100189006
2026-04	Out of State	82.5	SUPERPLAY LTD	December 2025	Company	100189011
2026-04	Out of State	215.86	CALLREVU LLC	December 2025	Company	100189273
2026-04	Out of State	1.24	NUGGLES	December 2025	Company	100189457
2026-04	Out of State	8.21	BEAUTY FROM BEES INC	December 2025	Company	100189459
2026-04	Out of State	2.35	AUTO POSHER	December 2025	Company	100190785
2026-04	Out of State	2.72	DC DIGITALS LLC	December 2025	Company	100192248
2026-04	Out of State	19	13LIONS MEDIA INC	December 2025	Company	100192335
2026-04	Out of State	86.75	INVENTIVE GARAGE & WORKSHOP	December 2025	Company	100193376
2025-12	Out of State	0.1	SPORTS ENDEAVORS INCORPORATED	December 2025	Company	100036424
2025-12	Out of State	0.06	MILLER'S INC	December 2025	Company	100122834
2025-12	Out of State	0.07	BOSCOV'S DEPARTMENT STORE LLC	December 2025	Company	100129168
2025-12	Out of State	0.09	JUST FOOD FOR DOGS LLC	December 2025	Company	100145440
2025-12	Out of State	0.03	TEXTMAGIC	December 2025	Company	100172079
2025-12	Out of State	0.03	TEAM DOG VIRTUAL	December 2025	Company	100176717
2025-12	Out of State	0.12	NATURAL STACKS INC	December 2025	Company	100183930
2026-03	Out of State	0.01	LENOVO (UNITED STATES) INC	December 2025	Company	010447437
2026-03	Out of State	-79	COMFORT MEDICAL LLC	December 2025	Company	010710597
2026-03	Out of State	-14	DIRECTV LLC	December 2025	Company	100119863
2026-03	Out of State	2.21	CURVY SENSE INC	December 2025	Company	100152790
2026-03	Out of State	0.11	HOLSTRENGTH LLC	December 2025	Company	100174846
2026-03	Out of State	3.34	TOTAL BULLSTRAP LLC	December 2025	Company	100181242
2026-03	Out of State	1.08	JUST CAUSE APPAREL	December 2025	Company	100184167
2026-03	Out of State	25	TROOMI WIRELESS INC	December 2025	Company	100185206

Figure 3.3: Combined Power BI sales tax table containing collection, filing, location, company, county, and revenue fields.

DAX measures and calculated columns were created to support total revenue, county summaries, revenue shares, missing-month indicators, company review status, and other analytical functions. DAX was also used to distinguish records requiring review from routine entries.

For example, a company-name review indicator was created to identify blank names, values that appeared to be dates, and generic entries such as unknown or null. The purpose was not to change the source automatically, but to create a transparent review queue through which staff could locate records requiring attention.

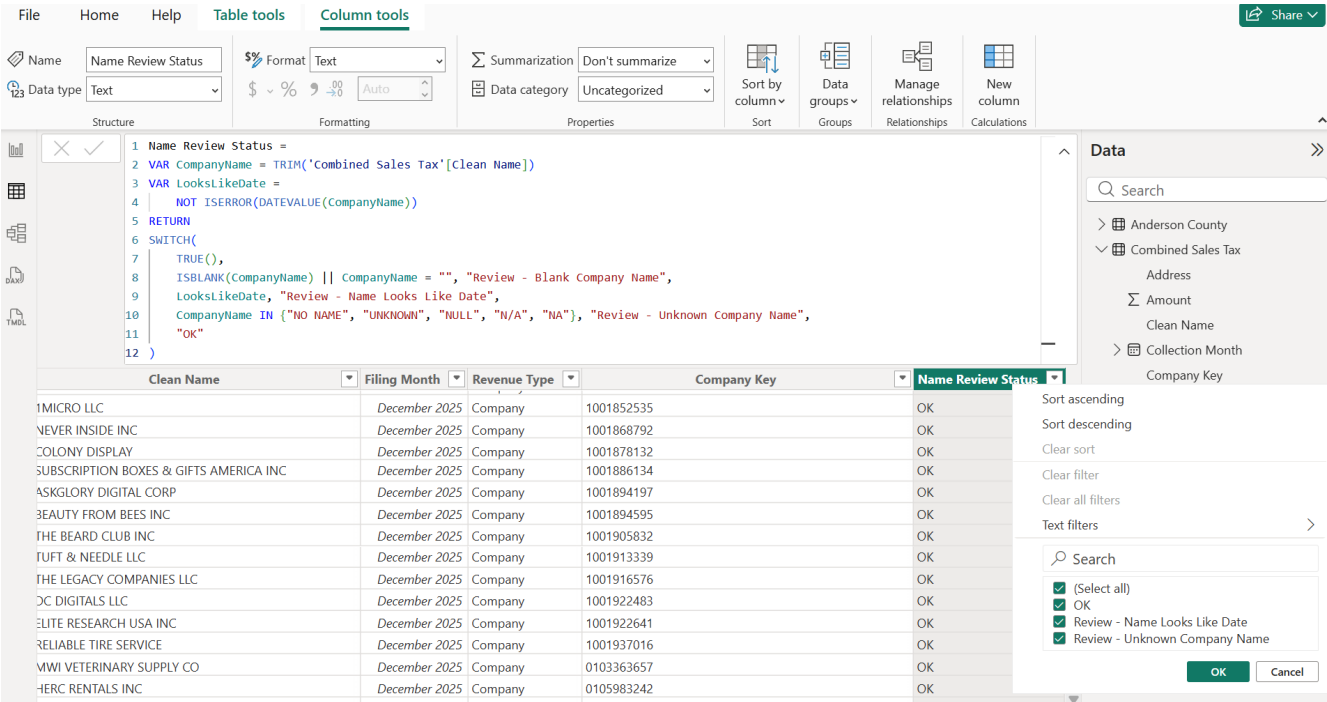


Figure 3.4: Example DAX logic used to create company-name data-quality review indicators.

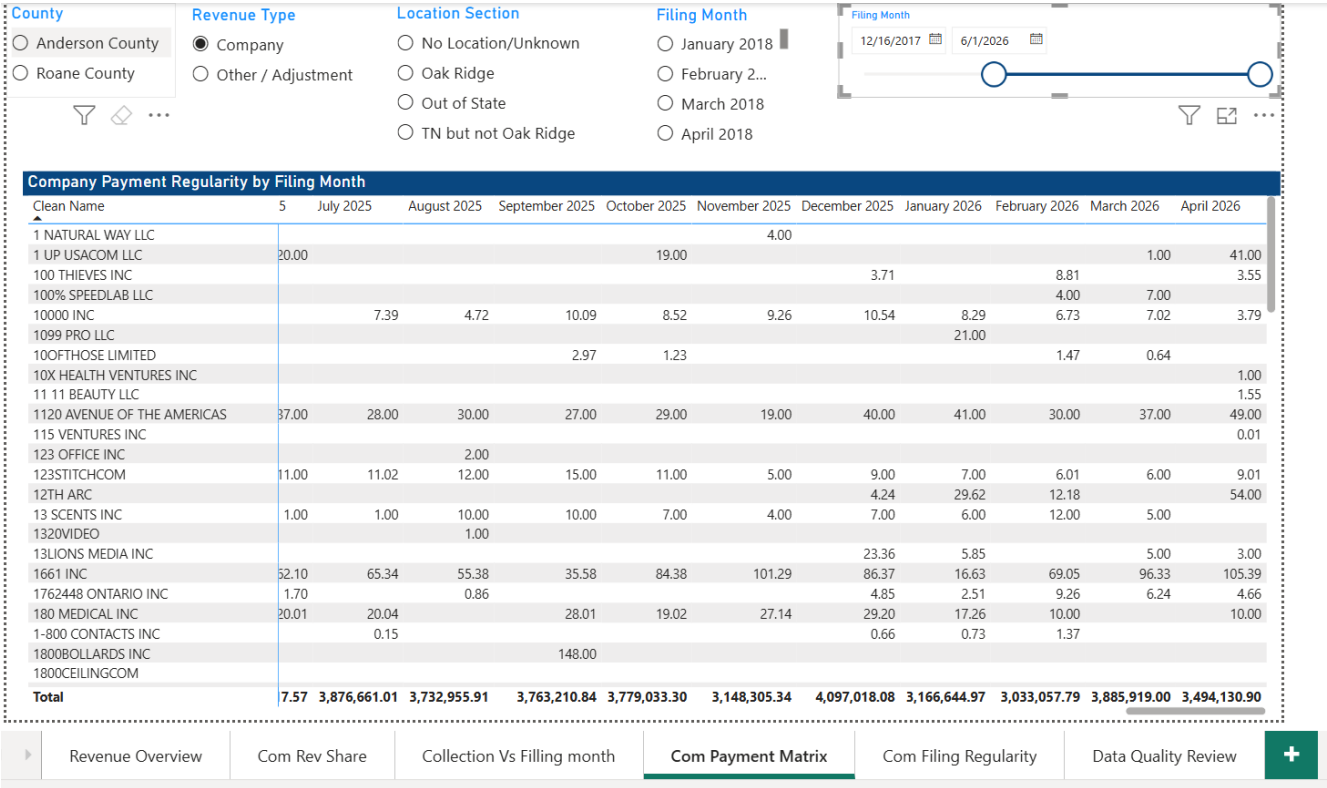


Figure 3.5: Company Payment Matrix.

The use of calculated review fields converted data-quality concerns into an operational feature of the system. Instead of discovering questionable records only during an ad hoc investigation, users could navigate directly to a review page and filter the dataset according to the type of issue identified.

The completed report was organized into separate analytical pages so that executive summaries, detailed company review, timing analysis, data-quality assessment, and forecasting could be performed without overcrowding a single dashboard. The principal pages are summarized in Table 3.2.

Table 3.2: Principal Power BI report pages and their functions

Report page	Principal function
Revenue Overview	Summarizes total revenue, county contributions, monthly trends, geographic distribution, revenue type, and the six-month forecast.
Company Revenue Share	Examines taxpayer concentration and the percentage of identified company revenue associated with individual businesses.
Collection versus Filing Month	Displays the relationship between the filing period associated with a payment and the month in which the revenue was collected.
Company Payment Matrix	Shows company-level payment amounts across months to identify regular, intermittent, missing, negative, or unusually large entries.
Company Filing Regularity	Identifies missing filing periods and provides indicators for records requiring further review.
Data Quality Review	Consolidates suspicious company names, unknown entries, unclear locations, missing identifiers, and related source-data concerns.
Revenue Forecast	Displays a six-month estimate derived from historical revenue collected by collection month, together with an uncertainty interval.

3.10 Revenue Overview Page

The Revenue Overview page, shown in Figure 3.6, was designed as the principal executive reporting view. It consolidated the most important revenue indicators into a single interactive page that could be understood quickly during a meeting or leadership discussion.

The page displayed total revenue collected, revenue by county, monthly revenue totals, monthly trends by county, revenue by taxpayer location and county, the division between company revenue and other or adjustment revenue, and a six-month forecast based on collection month.



Figure 3.6: Power BI Revenue Overview page combining summary, geographic, temporal, revenue-type, and forecasting information.

Three primary slicers allowed users to modify the page without changing its underlying data. The county slicer could display Anderson County, Roane County, or both. The revenue-type slicer separated identifiable company revenue from other or adjustment entries. The collection-month slicer allowed the reporting period to be narrowed or expanded.

The page followed a progression from broad summary to deeper interpretation. The total-revenue card answered how much revenue had been collected during the selected period. The county table showed how that total was divided between Anderson and Roane Counties. The monthly table and line chart showed how collections changed over time. The location chart indicated whether the associated businesses were located in Oak Ridge, elsewhere in Tennessee, outside the state, or lacked clear location information.

The revenue-mix chart showed the extent to which the total consisted of identified company payments rather than miscellaneous or adjustment records. The forecast visual added a prospective element by displaying a six-month baseline projection with an uncertainty range.

The page was intended to support conversation leading to decisions rather than merely display attractive graphics. A user could begin with a general question about total revenue and then use the county, date, and revenue-type filters to examine the source of an observed change within seconds.

3.11 Company Revenue Share Analysis

The Company Revenue Share page was developed to examine the distribution of revenue across taxpayers. Municipal sales tax collections can be concentrated among a limited number of major businesses, creating both fiscal strength and potential vulnerability. Understanding this concentration helps officials interpret monthly changes and identify

whether a large movement is broad-based or driven by one taxpayer.

The page calculated company revenue totals and each company’s share of total identified company revenue. Interactive filters allowed the analysis to be narrowed by county, collection period, filing period, location, and other available categories.

Company-level results require careful handling because taxpayer information and exact contributions may be sensitive even when portions of the underlying data are publicly accessible. For this reason, the final public presentation can describe the analytical capability and broad concentration patterns without displaying an identifiable top-taxpayer ranking or confidential company amounts.

The system nevertheless allows authorized internal users to examine these patterns when necessary. This is particularly useful when a large monthly increase or decrease appears in the overview page. Staff can move from the aggregate result to the company-level page and determine whether the change reflects several taxpayers, one unusually large payment, an adjustment, or a missing expected record.

3.12 Collection Month and Filing Month Analysis

One of the central analytical challenges was the difference between filing month and collection month. These fields could not be treated as interchangeable because the period associated with a taxpayer’s return might differ substantially from the period in which revenue appeared in the City’s collection records.

The Collection versus Filing Month page was designed as a matrix. Collection months appeared along one dimension, while filing months appeared across the other. Each cell represented revenue associated with a particular combination of filing and collection periods.

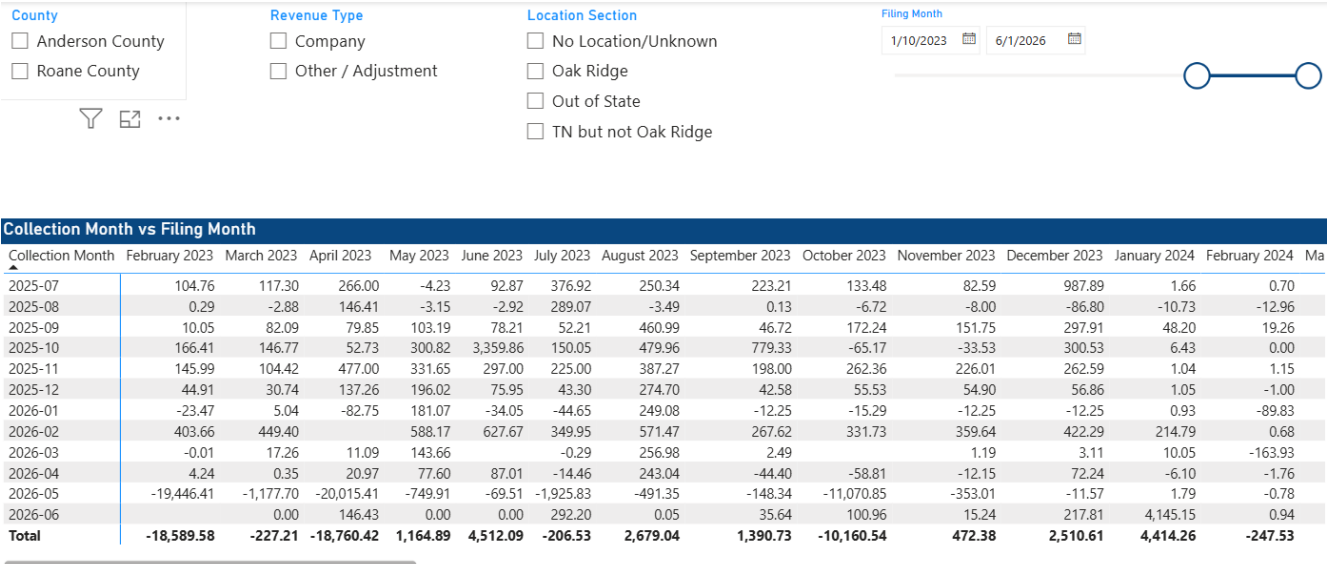


Figure 3.7: Collection-month and filing-month matrix used to examine timing differences and delayed records.

This structure made it possible to identify revenue collected in a current month that related to a substantially earlier filing period. It also revealed older filing dates that might otherwise be hidden inside a current aggregate total. Records extending back many years could be located and examined without manually searching each source file.

Timing differences may have several explanations, including delayed filings, corrections, amended returns, processing differences, or data-entry issues. The matrix does not determine which explanation applies. It provides an investigative structure through which unusual relationships become visible.

The page also supports filtering by county, revenue type, and location. This allows staff to determine whether timing differences are concentrated within one county, one class of records, or one type of taxpayer.

3.13 Company Payment Matrix

The Company Payment Matrix was developed to display company revenue across months in a format that made payment behavior easier to compare. Each company could be viewed across a sequence of filing or collection periods, allowing regular, intermittent, unusually large, negative, or absent payments to be identified visually.

The matrix was particularly useful because monthly totals alone can conceal company-level changes. A stable aggregate could occur even when one major taxpayer declined and another increased. Conversely, a sharp aggregate change might be explained by one delayed or unusually large payment.

The payment matrix allowed authorized users to examine whether a company appeared monthly, quarterly, annually, or irregularly. It also supported comparisons between expected payment frequency and observed filing behavior. The page did not assume that every company should appear each month. Instead, it provided the detailed history required to interpret the pattern in conjunction with the available frequency field.

3.14 Company Filing Regularity

The Company Filing Regularity page was designed to identify missing filing months based on the available historical record. A month-list table was used to compare expected periods with the months in which each company appeared.

The resulting measures identified the specific months absent from a company's history and counted the number of missing months. Additional statistics, including variation in monthly revenue, could help distinguish consistently small or intermittent businesses from companies with substantial but irregular activity.

The regularity page was intended as a review tool rather than a compliance determination. A missing month may be entirely consistent with a quarterly or annual filing schedule. Companies may also cease operations, change names, amend returns, or appear under a new registration identifier. The page therefore supports investigation when combined with filing frequency, company status, address, and other available information.

Figure 3.8 presents the filing-regularity review environment. The page combines observed filing history, missing-month calculations, filing-frequency information, and review indicators. The resulting flags are intended to direct staff attention toward unusual records; they do not independently establish that a taxpayer has failed to comply with a filing requirement.

County

Anderson County

Roane County

Location Section

No Location/Unknown

Oak Ridge

Out of State

TN but not Oak Ridge

Company Filing Gap Summary table

Clean Name	Missing Filing Months Since Jun-25	Number of Missing Months	Missing Months Flag	SD of Monthly Revenue Since 06/25
04/18/2022	2025-06, 2025-07, 2025-08, 2025-09, 2025-10, 2025-11, 2026-01, 2026-02, 2026-03, 2026-04, 2026-05, 2026-06	12	Review - More than 6 Missing Months	14.42
1 NATURAL WAY LLC	2025-06, 2025-07, 2025-08, 2025-09, 2025-10, 2025-12, 2026-01, 2026-02, 2026-03, 2026-04, 2026-05, 2026-06	12	Review - More than 6 Missing Months	1.11
1 UP USACOM LLC	2025-07, 2025-08, 2025-09, 2025-11, 2025-12, 2026-01, 2026-02, 2026-05, 2026-06	9	Review - More than 6 Missing Months	12.71
100 THIEVES INC	2025-06, 2025-07, 2025-08, 2025-09, 2025-10, 2025-11, 2026-01, 2026-03, 2026-05, 2026-06	10	Review - More than 6 Missing Months	2.65
100% SPEEDLAB LLC	2025-06, 2025-07, 2025-08, 2025-09, 2025-10, 2025-11, 2025-12, 2026-01, 2026-04, 2026-05, 2026-06	11	Review - More than 6 Missing Months	2.15
1099 PRO LLC	2025-06, 2025-07, 2025-08, 2025-09, 2025-10, 2025-11, 2025-12, 2026-02, 2026-03, 2026-04, 2026-05, 2026-06	12	Review - More than 6 Missing Months	5.82
100FTHOSE LIMITED	2025-06, 2025-07, 2025-08, 2025-11, 2025-12, 2026-01, 2026-04, 2026-05, 2026-06	9	Review - More than 6 Missing Months	0.91
10X HEALTH VENTURES INC	2025-06, 2025-07, 2025-08, 2025-09, 2025-10, 2025-11, 2025-12, 2026-01, 2026-02, 2026-03, 2026-05, 2026-06	12	Review - More than 6 Missing Months	0.28
11 11 BEAUTY LLC	2025-06, 2025-07, 2025-08, 2025-09, 2025-10, 2025-11, 2025-12, 2026-01, 2026-02, 2026-03, 2026-05, 2026-06	12	Review - More than 6 Missing Months	0.43
115 VENTURES INC	2025-06, 2025-07, 2025-08, 2025-09, 2025-10, 2025-11, 2025-12, 2026-01, 2026-02, 2026-03, 2026-05, 2026-06	12	Review - More than 6 Missing Months	0.00
Total	2026-06	1	OK	1,386,721.77

Figure 3.8: Company filing-regularity analysis and missing-month review indicators.

This feature created value beyond revenue reporting because it also revealed weaknesses in the administrative record. Users could identify where expected information was absent and determine whether further data collection, clarification, or registry review was required.

3.15 Data Quality Review

The Data Quality Review page consolidated records that met selected review conditions. These included blank company names, generic unknown entries, names that resembled dates, unclear locations, missing identifiers, and other records that could not be interpreted confidently.

This page separated analytical uncertainty from ordinary reporting. The Revenue Overview page displayed the complete fiscal picture, while the Data Quality Review page identified where the underlying information required additional attention. This approach avoided excluding uncertain records from totals while still making their limitations visible.

The system also enabled comparison between company names, registration identifiers, addresses, location classifications, business categories, filing history, and payment amounts. Through these comparisons, several types of anomaly were identified during the internship.

Some companies appeared to be registered at locations different from those expected based on their reported activity. Some registered businesses no longer appeared to exist under the reported name. Online businesses without a physical City location remitted small or infrequent amounts. Certain companies appeared to pay in months different from those anticipated, while others had substantial gaps in their available history.

A considerable amount of revenue also appeared under miscellaneous business or adjustment categories with

limited supporting information. Some large taxpayers were associated with business categories that appeared unusually small, while some known large companies generated less reported revenue than might have been expected. These observations did not establish error or noncompliance. They identified areas where registration, classification, timing, or payment information warranted additional verification.

The system therefore improved not only the use of existing data but also the City's understanding of what it did not yet know. Identifying missing or questionable information is an important analytical outcome because it informs future collection practices and helps officials avoid making decisions based on unexamined assumptions.

3.16 Revenue Forecasting

A six-month forecast was included on the Revenue Overview page using collection-month revenue. The forecast extended the system from retrospective reporting into prospective decision support.

The available detailed collection period covered twelve months, which limited the amount of historical information available for forecasting. Monthly revenue also contained fluctuations associated with large taxpayers, delayed filings, adjustments, and irregular activity. The forecast was therefore presented with an uncertainty band rather than as a precise prediction.

The principal value of the forecast was to create a baseline for discussion. It allowed officials to compare future actual collections with the pattern implied by the current record. As additional monthly files are incorporated, the historical series will grow and the model can be reassessed.

The forecast should not be interpreted as a guaranteed budget projection. It is sensitive to the period selected, the historical record available, and unusual payments. The system was designed so that users could understand this limitation visually through the projected range surrounding the central estimate.

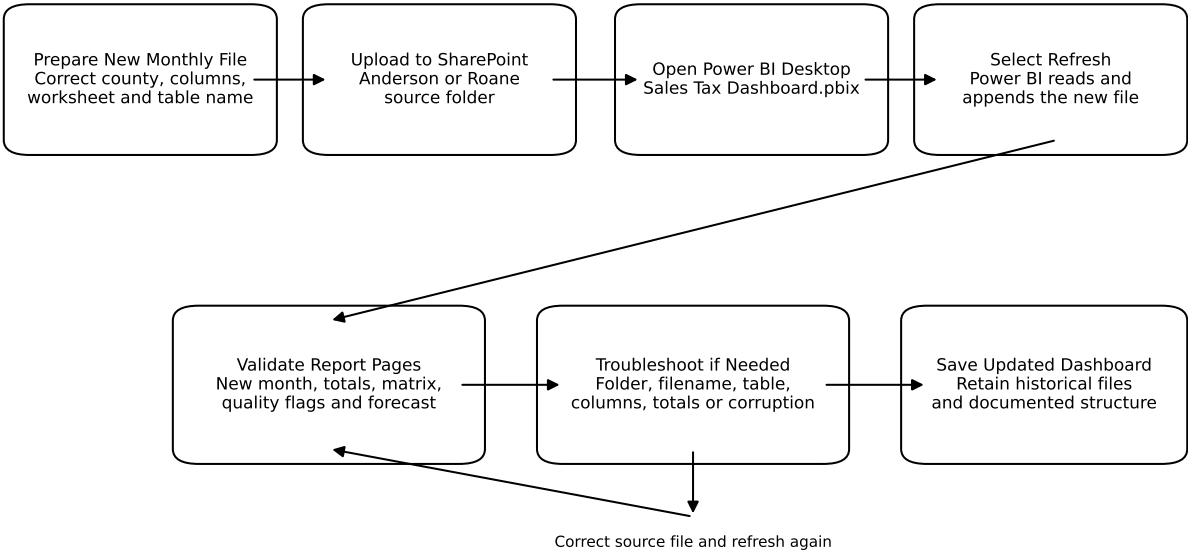
Future improvements could include seasonal models, comparisons with prior years once sufficient collection history is available, identification of major-payer effects, and external economic indicators. These extensions should be considered only after the underlying monthly series becomes longer and consistently maintained.

3.17 Monthly Update and Automation Process

The final workflow reduced the recurring monthly update to a small number of operational steps. When the new file follows the same structure as the existing state-provided records, staff ordinarily need only to place the file in the appropriate county SharePoint folder and refresh the Power BI report.

The operational sequence is illustrated in Figure 3.9. Under normal conditions, the user prepares the monthly file, uploads it to the appropriate county folder, refreshes the Power BI report, validates the updated pages, and saves the current dashboard. Troubleshooting is required only when the incoming source file does not follow the established structure.

SharePoint-Based Monthly Update and Validation Process



Under normal conditions, the recurring process consists of uploading the file and refreshing the dashboard.

Figure 3.9: SharePoint-based monthly update, refresh, validation, and troubleshooting process.

Anderson County files are stored in the Anderson County source folder, while Roane County files are stored in the Roane County source folder. The Power BI queries read all valid monthly files within those locations and append the new records to the historical model.

The automation depends on stable source-file requirements. These requirements are summarized in Table 3.3 and should be reviewed before each monthly upload.

Table 3.3: Monthly sales tax source-file requirements

Requirement	Required condition
County identification	Determine whether the incoming file contains Anderson County or Roane County records and upload it only to the corresponding folder.
File type	Use a valid Microsoft Excel workbook in .xlsx format when preparing files for the established automation.
File name	Use the approved county and month naming convention, such as AC_2026-06.xlsx or RC_2026-06.xlsx , consistent with the existing source files.
Worksheet name	The worksheet containing the transaction records must be named Data .
Excel table name	The transaction range must be formatted as an Excel table named exactly SalesTaxData .
Column structure	Column names and general structure must remain consistent with the historical monthly files unless the Power Query system is intentionally redesigned.
Summary rows	Total, grand-total, subtotal, or manually created summation rows must be removed before upload.
Historical files	Previous monthly files must remain in their county folders because Power BI uses the complete folder contents as its historical source.
Temporary files	Password-protected, corrupted, incomplete, or temporary files beginning with ~\$ must not be included in the source folder.

For Excel files, the worksheet containing the data must be named **Data**, and the table must be named **SalesTaxData**. Column headings must match the existing source structure. Total or summation rows must be removed so that Power BI does not import them as transactions.

Standard file names identify the county and reporting month. If the source arrives in another spreadsheet format, the records can be pasted into the required Excel structure and uploaded. Once the file is correctly prepared, the user opens the dashboard, selects refresh, waits for the queries to complete, and confirms that the new month appears on the relevant pages.

The update process then requires a series of validation checks. Staff confirm that the new collection month appears on the Revenue Overview page, the collection and filing matrix has updated, payment and regularity indicators reflect the new records, company revenue shares have changed where applicable, data-quality flags have been recalculated, and the forecast incorporates the latest month.

This design significantly reduced the technical burden of future analysis. There was no prior recurring process against which time savings could be measured because the files had been stored without regular analytical use. The benefit was therefore not merely that an existing process became faster. A previously unavailable analytical capability became operational through a two-step routine under normal conditions.

3.18 Documentation and Troubleshooting

Detailed instructions were prepared as an essential component of the final deliverable. Documentation included file naming, worksheet and table requirements, county-folder selection, SharePoint upload procedures, dashboard refresh steps, report-page validation, and common causes of failure.

Potential problems identified during development included uploading a file to the wrong county folder, using an incorrect file name, changing the worksheet name, changing the Excel table name, altering column headings, leaving a total row in the source, uploading a temporary or corrupted workbook, and modifying the SharePoint source path.

The documentation also explained why each condition mattered. Power BI depends on stable structural assumptions when combining multiple files. A changed worksheet or table name may prevent the query from locating the records. A changed column heading may interrupt a transformation step. A total row can inflate revenue by duplicating the summary of the transactions already present.

Users were instructed to correct the source file, replace the invalid version in the appropriate folder, reopen or refresh the dashboard, and verify each reporting page. They were also cautioned not to delete previous monthly files because the historical model depends on the full set of records remaining in the source folders.

The handover process included discussion with the supervisor regarding what might break, why it might break, how the failure would appear, and how it could be corrected. This knowledge transfer was as important as the dashboard itself. A reporting system that cannot be repaired after the original developer leaves does not provide durable institutional value.

3.19 Project Benefits

The system converted fragmented administrative records into a continuing municipal resource. Its benefits can be grouped into accessibility, analytical capacity, quality control, continuity, and strategic relevance.

Accessibility improved because users no longer needed to inspect multiple raw files individually to understand the revenue position. Summary indicators, tables, and charts were brought together within one interactive environment. County, date, and revenue-type filters allowed the same page to answer several questions without creating separate workbooks.

Analytical capacity improved because the City could examine monthly trends, geographic distribution, company shares, filing and collection timing, payment behavior, missing periods, data-quality issues, and forecast values through one connected model.

Quality control improved because unusual or incomplete records could be identified systematically. The system made missing information, unclear names, inconsistent locations, miscellaneous classifications, and irregular timing more visible. This supported better questions and more targeted follow-up.

Continuity improved because the workflow was designed for future use. The source-folder structure, automated refresh, written instructions, and troubleshooting guidance reduced dependence on the original developer. The system could continue accumulating data and become more informative over time.

The strategic value of the system extends beyond financial reporting. Municipalities compete for investment, business growth, infrastructure opportunities, and fiscal stability. A city that can demonstrate reliable information,

identify changes quickly, and understand its own revenue base is better positioned to evaluate opportunities and communicate with stakeholders.

The system therefore helped transform data already available to the City into a form capable of supporting discussion and decision-making. Its most important contribution was not one particular graph or calculation. It was the creation of a framework through which Oak Ridge could continue using, questioning, and improving its sales tax information after the internship concluded.

3.20 Relevance to Other Tennessee Municipalities

The broader relevance of the project is particularly important for the IPS and MTAS audience. Tennessee municipalities may receive sales tax and administrative information through similar state systems, spreadsheet files, clerical reports, or local registries. Many may possess valuable records without having an established method for integrating them into recurring analysis.

The exact Oak Ridge model cannot be transferred automatically to every municipality because local file structures, administrative practices, software access, and reporting priorities differ. The general architecture, however, is replicable.

A municipality can establish standardized source-file requirements, separate folders for recurring records, repeatable Power Query transformations, a combined analytical table, interactive Power BI pages, quality-control indicators, and documented monthly update procedures. The specific measures and visuals can then be adapted to local questions.

The project demonstrates that local governments do not necessarily need to begin with an expensive external platform or an extensive data-science department. A carefully designed system using tools already available within the Microsoft environment can create substantial analytical value when it is aligned with institutional capacity and maintained consistently.

The most important requirement is not software alone. Successful replication depends on understanding the source data, consulting the employees who use it, documenting assumptions, preserving confidentiality, designing for future staff, and treating data quality as an ongoing process rather than a one-time cleaning exercise.

3.21 Limitations and Recommendations

The system has several limitations that should be recognized. The detailed collection history currently covers only twelve months, limiting long-term trend analysis and the reliability of forecasting. As additional monthly records are incorporated, more meaningful seasonal and year-over-year comparisons will become possible.

The historical filing record extends much further, but older fields may contain greater inconsistency and incomplete descriptive information. Historical entries should not be assumed to have the same level of quality or comparability as newer records without further review.

The system also depends on stable source structures. Changes made by the state or another data provider to column names, worksheet formats, file types, or field definitions may interrupt the automated queries. Future users should compare each new file with the existing structure before uploading it.

Company identity remains a continuing challenge. Businesses can change names, addresses, registration identifiers,

ownership, or operational status. A separate maintained company reference table could improve matching across periods and preserve relationships among former names, current names, identifiers, and locations.

Miscellaneous and adjustment revenue should receive continued attention. These values are included appropriately in total revenue, but limited descriptions reduce the City's ability to interpret their sources. Additional information from the relevant administrative provider could improve classification.

The location categories could also be strengthened through systematic address validation or geocoding, subject to confidentiality and data-access requirements. This would allow more precise distinctions among companies physically located in Oak Ridge, registered elsewhere, or operating through online transactions.

The forecast should be treated as preliminary until a longer collection history is accumulated. Future modeling could consider seasonality, taxpayer concentration, economic conditions, and alternative forecast methods. Any expansion should retain transparent assumptions and avoid presenting uncertain values as guaranteed outcomes.

The City should continue maintaining both the Excel and Power BI versions during the early period after handover. This provides flexibility for users with different preferences and offers an alternative reference environment if one platform experiences an issue.

Finally, the system should be reviewed periodically rather than treated as permanently complete. New questions, data fields, users, and administrative requirements will emerge. Its future value will depend on the same principles that guided its creation: accessibility, replicability, maintainability, scalability, and safe analytical exploration.

Davenport Energy-Sector Revenue Projection Project

4.1 Project Background

The second major internship assignment focused on estimating the potential fiscal effects of future energy-sector investment in Oak Ridge. Unlike the sales tax project, which analyzed historical administrative records, this assignment addressed future municipal revenue associated with proposed industrial development. The work was conducted using the Davenport fiscal impact framework, a planning methodology commonly employed to estimate the public revenue generated by large development projects.

Rapid expansion within the energy sector has created significant opportunities for Oak Ridge. New investment associated with advanced manufacturing, nuclear technology, enrichment facilities, and related infrastructure has the potential to alter the City's long-term fiscal position. Local officials therefore required a structured method for estimating the property tax revenue that future projects could generate under different investment assumptions.

The assignment concentrated on projected revenue rather than historical collections. Because many projects had not yet been completed, traditional statistical forecasting based on historical tax collections was not appropriate. Instead, projected investment values, assessment procedures, depreciation assumptions, and local tax rates were combined within the Davenport framework to estimate future municipal revenue under alternative scenarios.

The objective was not to predict exactly how much revenue the City would receive. Rather, it was to develop a transparent analytical model that could evaluate different investment assumptions, communicate the effects of policy choices, and support long-term financial planning.

4.2 Property Tax and Industrial Investment

Property tax represents one of the principal revenue sources available to local governments. Industrial investment affects municipal finances because new facilities increase the assessed value of taxable property. As facilities are constructed, expanded, or modernized, the taxable assessment changes according to state assessment rules, depreciation schedules, and any applicable economic development incentives.

Large energy projects often involve investments worth hundreds of millions or even billions of dollars. These investments may include buildings, specialized equipment, production facilities, transmission infrastructure, and supporting improvements. Because industrial equipment depreciates over time and incentive agreements frequently reduce taxable values during initial years, the relationship between investment and municipal revenue is considerably more complex than applying a single tax rate.

Reliable projections therefore require an analytical framework capable of incorporating investment timing, equipment life, depreciation schedules, assessment ratios, local tax rates, and incentive structures simultaneously.

4.3 The Davenport Fiscal Impact Framework

The Davenport model provides a structured method for estimating future local tax revenue generated by industrial investment. Rather than relying on a single estimate, the framework evaluates how investment evolves over time as property values change and equipment depreciates.

The model follows the sequence through which investment becomes municipal revenue. Capital investment is first allocated between different categories of taxable property. Assessment ratios are then applied according to Tennessee property assessment procedures. Depreciation schedules reduce taxable values over time, after which local property tax rates determine annual tax revenue.

The framework allows different assumptions to be modified independently. Investment amounts, construction timing, equipment life, depreciation rates, assessment ratios, tax rates, and incentive agreements can all be adjusted without rebuilding the entire model. This flexibility makes the approach particularly useful for municipal planning because officials frequently evaluate several possible development scenarios before major projects are completed.

Throughout the internship, the Davenport framework was used primarily as a planning tool rather than a forecasting model in the statistical sense. Its purpose was to estimate potential fiscal outcomes under specified assumptions instead of predicting future market behavior.

4.4 Data Collection

Developing the projection model required collecting information from several municipal and external sources. Available documentation included investment estimates, property classifications, assessment guidance, depreciation assumptions, tax rates, economic development materials, and publicly available information concerning proposed energy-sector facilities.

The information available for different projects varied considerably. Some developments contained detailed investment estimates and construction schedules, while others included only preliminary announcements or broad planning documents. Consequently, the model was designed to accommodate different levels of information availability without requiring identical inputs for every project.

Considerable effort was devoted to understanding how different assumptions interacted within the model. Discussions with municipal staff provided important institutional context regarding assessment procedures, local development incentives, and practical considerations affecting industrial investment.

4.5 Model Development

The projection model was developed in Microsoft Excel to provide transparency, flexibility, and ease of future maintenance. Individual worksheets were organized according to major components of the Davenport methodology, including investment assumptions, assessment calculations, depreciation schedules, tax calculations, and projected annual revenue.

Each section of the model was designed so that changing one assumption automatically updated all dependent calculations. This structure reduced manual recalculation while allowing multiple investment scenarios to be evaluated efficiently.

Inputs were clearly separated from calculated values to improve transparency and reduce the likelihood of accidental modification. Formula structures were documented carefully so that future users could understand how projected revenue was generated without reconstructing the underlying calculations.

The model emphasized clarity over unnecessary complexity. Municipal planning tools must remain understandable to future staff who may not possess specialized programming backgrounds. Accordingly, calculations were organized in a manner that balanced analytical rigor with practical usability.

4.6 Application to Municipal Decision Making

The projection model supports several aspects of municipal planning. First, it provides estimates of future property tax revenue that can inform long-term budgeting discussions. Second, it allows officials to compare alternative development proposals using a consistent analytical framework. Third, it improves communication by presenting complex financial calculations in a structured and transparent manner.

The model may also support infrastructure planning. Large industrial developments often require additional public investment in transportation, utilities, emergency services, and other municipal infrastructure. Understanding the timing and magnitude of future tax revenue helps officials evaluate whether anticipated revenue is sufficient to support these investments.

The framework additionally contributes to discussions regarding economic development incentives. Because incentive agreements influence taxable values during early project years, the model helps illustrate the tradeoffs between short-term tax reductions and potential long-term economic benefits.

4.7 Project Limitations

Several limitations should be recognized when interpreting the projection results. Future investment values remain uncertain because construction schedules, project costs, market conditions, and regulatory environments may change after initial planning documents are prepared.

Depreciation assumptions and equipment lives represent generalized estimates rather than precise predictions for every individual asset. Likewise, future tax rates and assessment policies may change over time through legislative or administrative action.

Accordingly, the projected revenue estimates should be interpreted as planning scenarios based upon available information and stated assumptions rather than guaranteed future outcomes.

These limitations do not diminish the usefulness of the model. Instead, they emphasize the importance of periodically updating assumptions as new information becomes available.

The pictures demonstrate the organization of model inputs, formula-based projections, PivotTable summaries, and long-term visual outputs. Company names, project-specific assumptions, and confidential financial values are not disclosed.

development.

Together, the two projects illustrate the complementary roles of retrospective and prospective analysis within municipal finance. One improves understanding of existing fiscal conditions, while the other supports planning for future opportunities.

The completed projection framework provides the City with a transparent analytical tool that can be expanded as additional investment information becomes available. Future staff can update assumptions, evaluate new development proposals, and compare alternative scenarios without rebuilding the model from the beginning.

The project therefore contributes to the City's long-term analytical capacity by transforming future investment information into a structured decision-support framework that can assist budgeting, strategic planning, and economic development discussions.

Professional Development and Reflection

5.1 Learning Local Government from the Inside

Before beginning this internship, my understanding of local government was largely limited to the public services that residents encounter in everyday life. Like many citizens, I viewed municipal government primarily as an organization responsible for maintaining roads, providing basic administrative services, and collecting taxes. Working within the City of Oak Ridge fundamentally changed that perspective by exposing me to the extensive planning, coordination, financial analysis, and operational responsibilities required to support an entire community.

Throughout the internship I observed that municipal government functions as a highly interconnected organization in which decisions made within one department frequently influence the work of many others. Financial planning affects infrastructure investment, emergency services influence development planning, engineering supports economic growth, and administrative decisions require continuous coordination among departments. This interconnected structure demonstrated that effective local government depends not only on technical expertise but also on communication, collaboration, and long-term planning.

One of the most surprising aspects of the experience was the breadth of services provided directly by the City of Oak Ridge. The City manages an unusually wide range of municipal functions despite its unique historical role within the national nuclear enterprise. Learning how these responsibilities are coordinated greatly expanded my appreciation for municipal administration and public service.

5.2 Working Alongside Public Servants

Perhaps the most valuable aspect of the internship was the opportunity to work closely with professionals whose careers are dedicated to public service. Throughout the internship I interacted with staff members from multiple departments whose commitment to serving the community consistently exceeded my expectations.

Particularly memorable were my interactions with emergency management personnel, police officers, firefighters, and other municipal employees whose responsibilities require constant readiness and rapid decision making. Despite demanding workloads, these individuals remained approachable, patient, and genuinely interested in explaining their work to interns. Their professionalism demonstrated that effective public service combines technical competence with empathy, teamwork, and a sincere commitment to the community.

Observing these departments reinforced that local government operates successfully because of dedicated professionals whose daily efforts often remain invisible to the public. Their willingness to share their knowledge made the internship both educational and personally meaningful.

5.3 Exposure to Municipal Leadership

The internship also provided opportunities to observe municipal leadership at several organizational levels. Meetings with department directors, senior administrators, and elected officials demonstrated how technical analysis supports

public decision making.

Among these experiences, attending the Oak Ridge City Council meeting was particularly memorable. Seeing council members discuss policy, review public information, consider staff recommendations, and make decisions affecting residents illustrated how analytical work ultimately contributes to democratic governance. Observing the complete decision-making process connected the technical components of my internship with their broader public purpose.

Equally valuable were numerous meetings held throughout the internship in which staff discussed ongoing projects, development initiatives, operational challenges, and strategic planning efforts. These conversations highlighted the importance of presenting information clearly so that technical analysis can effectively inform policy decisions.

5.4 Growth as an International Student

As an international student, participating in this internship carried additional significance beyond professional development. The opportunity provided by the University of Tennessee Institute for Public Service to work directly within a municipal government allowed me to gain practical experience that would otherwise have been difficult to obtain.

Previous international interns have also participated in similar placements, demonstrating the Institute's commitment to creating meaningful opportunities for students from diverse backgrounds. By integrating international students into local government operations, the Institute not only supports professional development but also encourages cultural exchange, knowledge sharing, and greater understanding of public administration in the United States.

For me personally, the experience strengthened my understanding of American local government while allowing me to contribute analytical work that supported ongoing municipal initiatives. Being trusted with substantive assignments reinforced my confidence as a researcher and demonstrated that meaningful contributions can come from individuals with diverse educational and cultural backgrounds.

5.5 Professional Growth

The internship significantly strengthened both my technical and professional skills. From a technical perspective, I gained substantial experience organizing large administrative datasets, developing automated reporting systems, creating interactive Power BI dashboards, validating data quality, constructing analytical workflows, and developing long-term fiscal projection models.

Equally important were the professional skills developed throughout the internship. Regular interaction with supervisors, department directors, and municipal staff improved my ability to communicate technical information to nontechnical audiences. Working on projects intended for practical decision making reinforced the importance of documentation, transparency, accuracy, and maintainability in analytical work.

The experience also emphasized that successful public-sector analysis requires balancing methodological rigor with practical usability. Analytical tools are valuable only if future employees can understand, maintain, and apply them effectively. This principle influenced both major projects completed during the internship.

5.6 Connecting Research with Public Service

My graduate education emphasized econometrics, quantitative analysis, and economic research. The internship demonstrated how these academic skills can directly support public administration when applied to real municipal problems.

The sales tax reporting system required transforming fragmented administrative records into actionable information that could improve operational efficiency. The Davenport revenue projection model required applying economic reasoning to evaluate long-term fiscal implications of industrial investment. Both assignments illustrated that quantitative methods are most valuable when they help decision makers better understand complex policy questions.

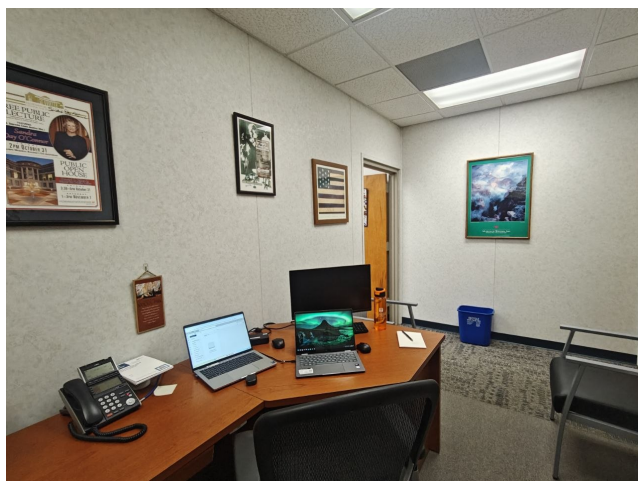
The internship therefore strengthened my interest in applying analytical methods within public institutions, particularly in areas involving economic development, municipal finance, policy analysis, and data-driven decision making.

The internship also provided an opportunity to observe municipal decision making through attendance at an Oak Ridge City Council meeting. As shown in Figure 5.1, the experience offered a direct view of how elected officials, professional staff, and community members participate in local governance and public deliberation.



Figure 5.1: Attendance at an Oak Ridge City Council meeting during the internship.

Figures 5.2a and 5.2b document the professional setting in which the internship was completed. Beyond the technical assignments, the placement provided regular opportunities to work within the municipal environment, interact with public-service professionals, and better understand the daily operations of local government.



(a) Professional activities during the City of Oak Ridge internship.



(b) Internship experience within the municipal work environment.

Figure 5.2: Selected photographs from the IPS/MTAS internship placement with the City of Oak Ridge.

5.7 A Lasting Perspective

Looking back, the internship provided far more than technical experience. It fundamentally changed my understanding of how local governments operate and the role that analytical work plays in serving communities.

I leave the internship with a much deeper appreciation for the dedication of public servants, the complexity of municipal administration, and the importance of reliable information in effective governance. Equally important, I leave with confidence that the analytical skills developed during my graduate education can contribute meaningfully to solving real public-sector problems.

The opportunity to work alongside professionals at the City of Oak Ridge and the University of Tennessee Institute for Public Service has been one of the most valuable experiences of my academic and professional journey. The lessons learned throughout these two months will continue to shape both my future research interests and my professional career.

Conclusion

6.1 Summary of Internship Outcomes

This internship provided an opportunity to apply graduate-level quantitative training to practical problems within municipal government. During the two-month placement with the City of Oak Ridge through the University of Tennessee Institute for Public Service, I completed two major analytical assignments that addressed different aspects of municipal financial planning. The first project developed an automated reporting system for local sales tax analysis by integrating historical administrative records into a standardized database and interactive Power BI dashboard. The second project developed a long-term fiscal projection framework using the Davenport methodology to estimate future property tax revenue associated with energy-sector investment.

Although the projects differed substantially in methodology and purpose, both were designed to improve the City's analytical capacity. The sales tax project strengthened the City's ability to understand historical and current revenue patterns through automated reporting, data validation, and visualization. The Davenport project strengthened long-term planning by providing a structured framework for evaluating future development scenarios under varying economic assumptions. Together, the projects demonstrate how data management, quantitative analysis, and transparent modeling can support evidence-based municipal decision making.

Beyond the technical deliverables, the internship emphasized the importance of designing analytical systems that remain useful after the original developer is no longer involved. Documentation, automation, transparency, and ease of maintenance were incorporated throughout both projects to ensure that future staff members can continue updating, expanding, and applying the systems with minimal manual effort.

6.2 Professional Reflection

Working within municipal government fundamentally expanded my understanding of public administration. The internship demonstrated that effective local government depends not only on sound policy but also on reliable information, collaboration across departments, and careful long-term planning. Observing how analytical work contributes to decisions affecting residents reinforced the practical value of economic analysis outside traditional academic settings.

The experience also strengthened my appreciation for public servants whose work often occurs behind the scenes. Throughout the internship I observed professionals who approached their responsibilities with dedication, accountability, and a genuine commitment to serving the community. Their willingness to share knowledge and involve interns in meaningful projects created an exceptional learning environment and significantly enriched my professional development.

As an international student, I am especially grateful for the opportunity provided through the University of Tennessee Institute for Public Service. The internship allowed me to gain firsthand experience within American local government while contributing analytical work to projects with practical value. Such opportunities not only strengthen students professionally but also encourage meaningful collaboration between universities and public

institutions.

6.3 Future Opportunities

The systems developed during this internship provide a foundation for future analytical work rather than representing completed endpoints. The sales tax reporting system can continue expanding as additional historical records become available, while future dashboard enhancements may incorporate more advanced performance metrics, geographic analysis, or automated anomaly detection.

Similarly, the Davenport projection framework can be updated as new information becomes available regarding industrial investments, assessment policies, depreciation schedules, and local economic development initiatives. Because both systems were designed with flexibility and maintainability in mind, they can continue evolving alongside the City's future analytical needs.

More broadly, this internship demonstrated the value of combining quantitative analysis with practical public administration. As municipalities increasingly rely on data to support policy development and financial planning, investments in analytical capacity will become increasingly important. Systems that transform administrative records into accessible information allow local governments to make more informed decisions while improving organizational efficiency and transparency.

6.4 Final Remarks

This internship has been one of the most meaningful experiences of my graduate education. It provided an opportunity to apply classroom knowledge to real municipal challenges while learning directly from experienced public servants committed to improving their community. The projects completed during the internship strengthened my technical skills, expanded my understanding of local government, and reinforced my interest in applying quantitative methods to public policy and economic development.

I am sincerely grateful to the City of Oak Ridge for welcoming me into its organization and entrusting me with meaningful analytical responsibilities. I am equally grateful to the University of Tennessee Institute for Public Service and the Municipal Technical Advisory Service for creating an internship program that allows students to contribute directly to local government while gaining valuable professional experience. The guidance provided by Deputy City Manager Tom Pessemier, along with the support of the Institute for Public Service staff, made this internship both educational and professionally rewarding.

The lessons learned throughout these two months will continue to influence my future academic work and professional career. I hope that the analytical systems developed during this internship will continue supporting municipal decision making and contribute, even in a small way, to the continued growth and success of the City of Oak Ridge.

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